

Commvault Professional

Course Guide

Module 4 – Servers and Features

Commvault Platform Release 2023E

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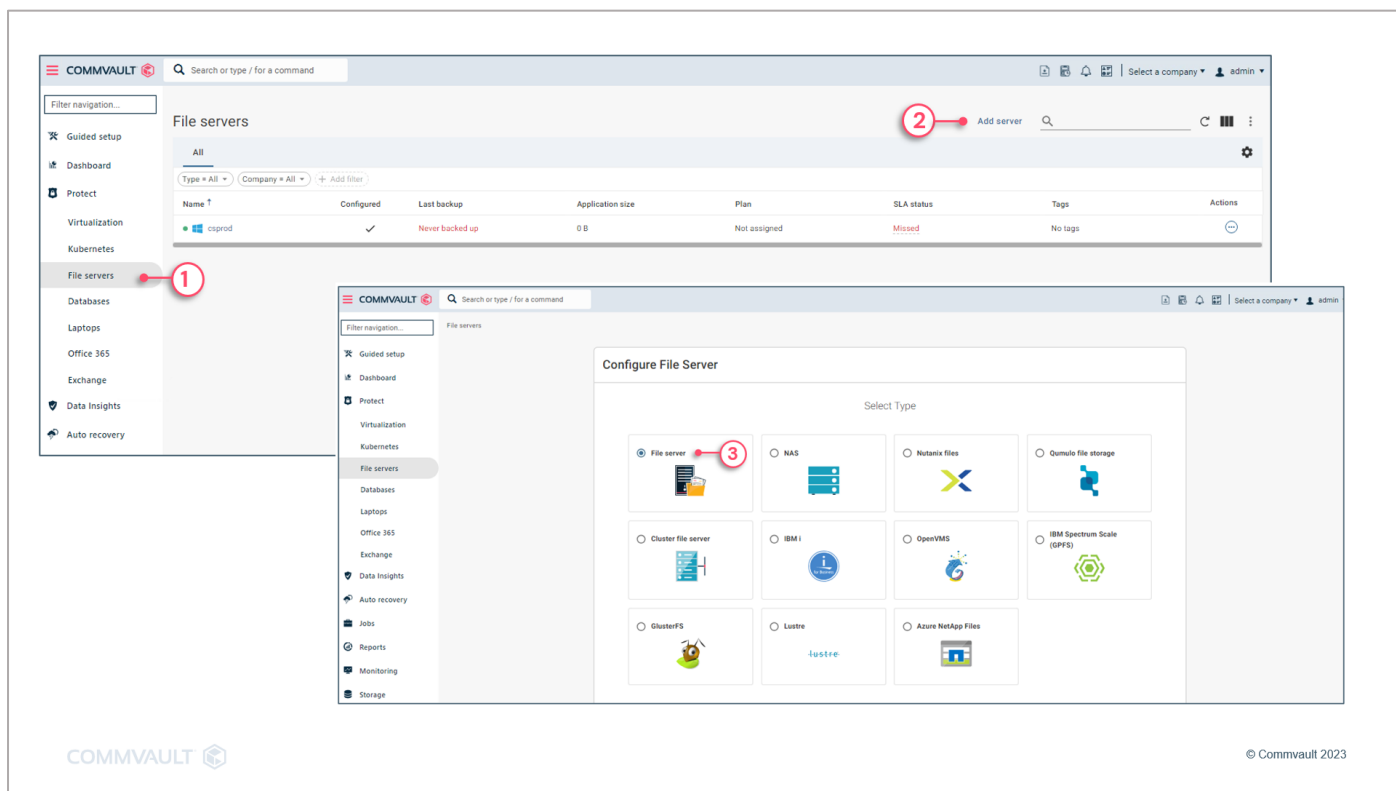
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Virtual Machines Protection 33

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Agent Deployment

Push Install



Use the Command Center to conduct a “push install” to servers using the specific ports.

- **Commvault base services** –8400 and 8401
- **Linux** – 22
- **Windows** — 135, 139, and 445

Execute a Push Install

1. Select **Protect > File servers**
2. Click **Add server**
3. Select the type of file server to configure; for a Windows or Linux server, select **File server**

Push Install ...cont'd

Add File Server

Server Configuration

1 Server Configuration
2 Plan
3 Backup Content

Server Configuration

Name * 4

Host name

User name 5

Password 5

Confirm password 5

OS Type
☒ Windows 6
☐ Unix and Linux

Installation location
Enter the installation location (Optional) 7

Reboot if required ☐

To install the software interactively, click here

Plan

Configure plan ☒

Search plans by plan name + C

Plan	RPO	Recovery Time	Primary storage type	Associated entities
Production Server plan	1 day	2	Disk	1

8

Backup Content

Backup content ADD 9

Exclusions ADD 9

Include global exclusions ☐

Use cell level policy ☐

☒ Backup system state

EQUIVALENT API

10

4. Enter the server names or IP addresses
5. Enter credentials with administrative privileges to install software on server
6. Choose the platform
7. Enter a new location if you do not want to install the software in the default directory
8. Choose a server plan
9. Click **Add** to override the default content definition
10. Click **ADD**

Push Install ...cont'd

The screenshot shows the Commvault Jobs page. The left sidebar has a 'Jobs' menu item highlighted with a red circle labeled '11'. The main area displays a table of active jobs. The table has columns: Job id, Operation, Status, Server, Agent type, Subclient, Size, Start, Elapsed, Progress, Error description, and Actions. A red circle labeled '12' points to the 'Error description' column header. A red circle labeled '13' points to the 'Job id' column header. The table shows one job: Job 34, Install Software, Running, on server dc1.cv.lab, with a progress bar at 0%.

Below the main table, there is a detailed view for 'Job 34 - [Install Software]'. This view includes sections for General, Associations, Client status, Progress, and Item status.

General

Property	Value
Type	Install Software
Status	Running
Job started by	admin (Started interactively)

Associations

Property	Value
Source client computer	dc1.cv.lab

Client status

Property	Value
Total clients	1
Clients already up-to-date	0
Successful clients	0
Clients partially successful	0
Skipped clients	0
Pending clients	0
Failed clients	0

Progress

Property	Value
Start time	Sep 8, 2023, 6:51:49 AM
Current phase	Install Client
Elapsed time	51 seconds
Last update time	Sep 8, 2023, 6:51:50 AM
Current throughput	N/A

Item status

Property	Value
Number of files transferred	0

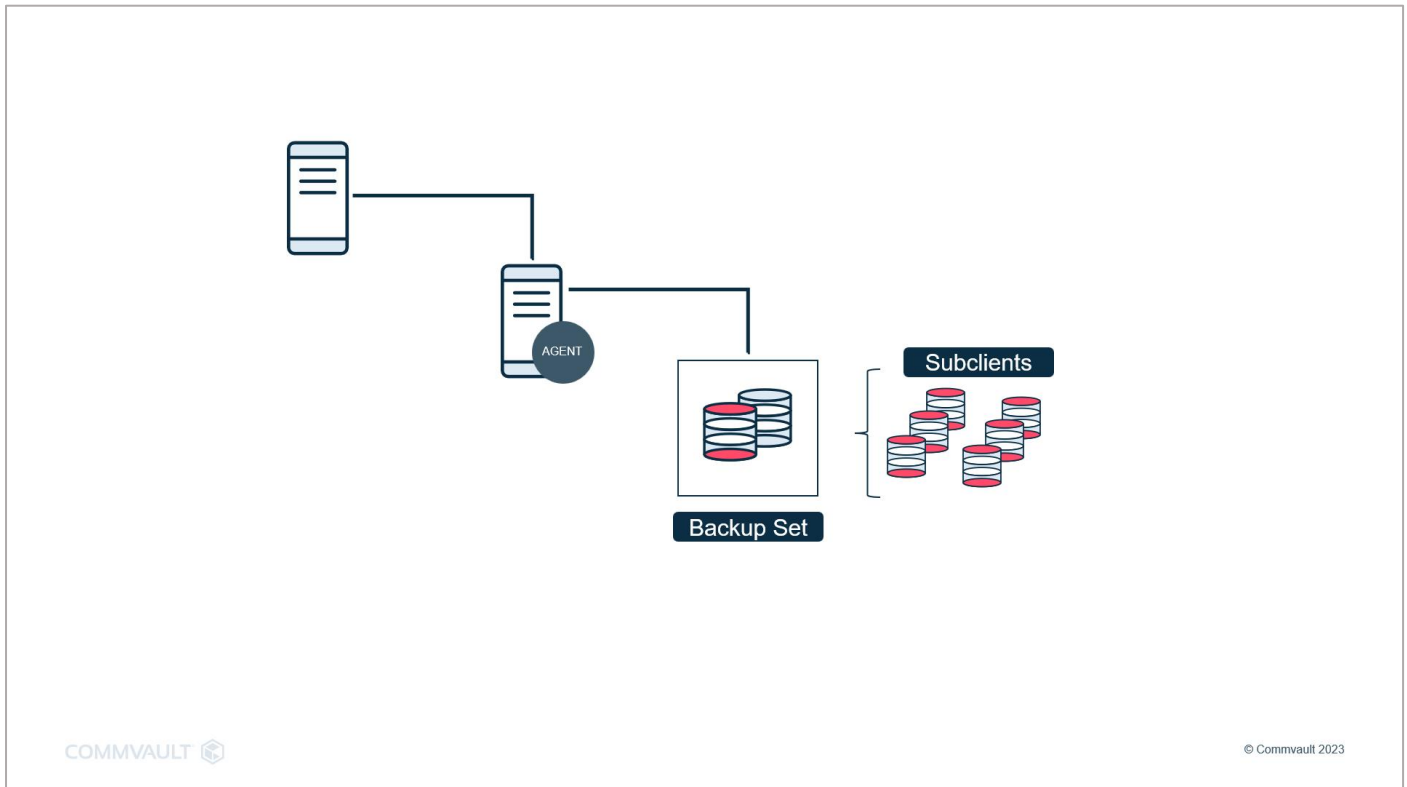
11. Select **Jobs**

12. The progress of the installation is displayed

13. Click the **Job id** for detailed information

View the details of the job, including the error description.

Server Administration



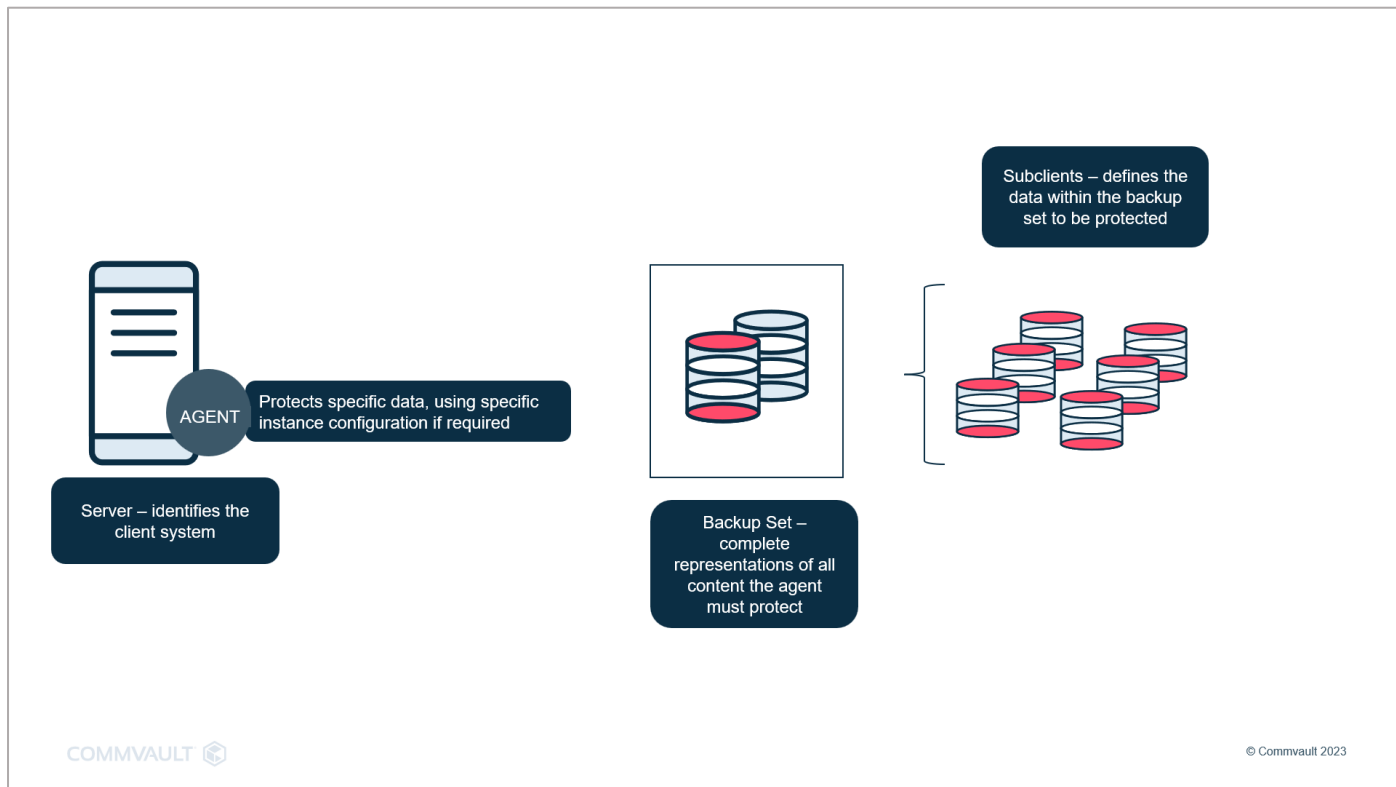
Server Administration

Commvault software uses agents to communicate with file systems and applications that require protection. Any server with an agent installed on it is referred to as a “server.” Each agent contains code that is used to communicate directly with the system requiring protection. The agent communicates using Application Program Interfaces (APIs) or scripts that are native to the file system or application.

For example, a Windows 2019 file system uses Volume Shadow Service (VSS) to protect file data, so the Windows agent has the option to enable VSS during backup operations. The agent then has a backup set defined. The backup set is a complete representation of all data the agent is responsible for protecting. Within the backup set, subclients are used to define the actual data requiring protection. By default, a default subclient is used to define all data requiring protection within the backup set.

Additional subclients can be created to define specific content requiring protection. When content is defined within the user defined subclient, it automatically is excluded from the default subclient. An example for a custom subclient could be defining a specific drive containing user data where VSS is initiated for the drive during backup jobs to ensure all open files are protected.

Server Navigation Structure



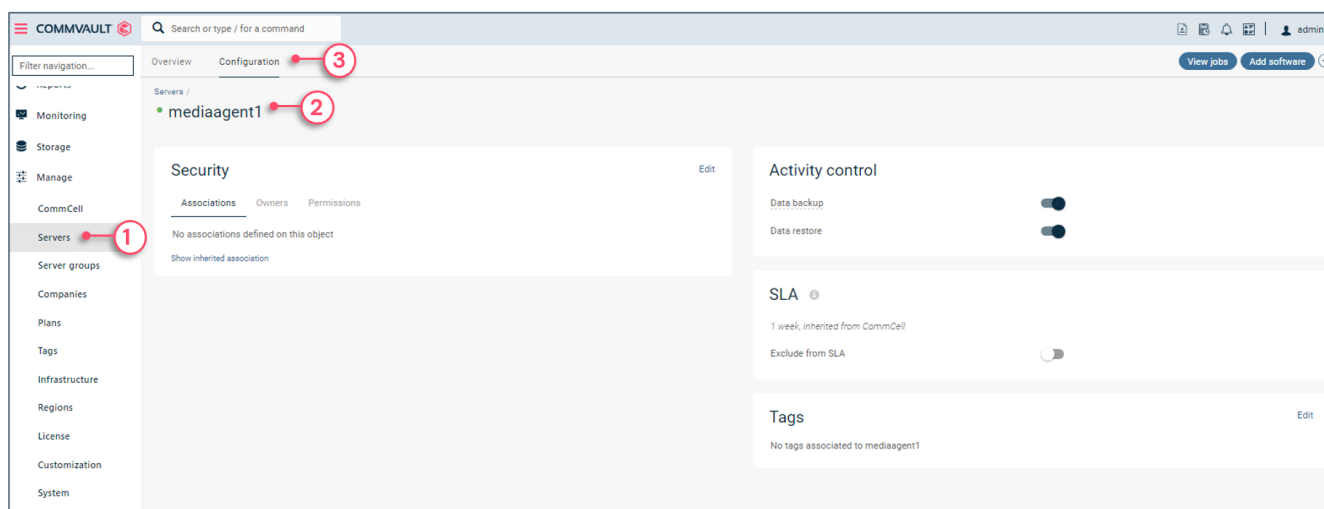
Server Navigation Structure

The Server navigation structure defines a hierarchy for arranging various components of a server. This hierarchy remains consistent throughout servers with different agents installed.

Each component in the server navigation structure provides specific functionality to properly configure, manage, and protect production data. Each of these components has specific features and properties that are configured to provide a comprehensive protection strategy.

- **Server** — Identifies the client system by CommCell server name and Host name or IP address
- **Agent** — Installed on the server or access node to protect specific data such as files, databases, emails, and virtual machines
- **Instance** — Used with specific types of agents such as database application or hypervisors to define configuration
- **Backup Set** — Provides a complete representation of all content the agent is responsible to protect
- **Subclient** — Defines the actual data within the backup set that requires protection

Server Properties



Access Server Properties

Few options and parameters are available for servers. These options are applied globally to any agent types installed on the server. Options are grouped by security, agents, and activity control.

1. Select **Manage > Servers**
2. All servers are displayed in the view select the server name to open its properties
3. Select the *Configuration* tab. Client options are divided into sections

Check Server Connectivity

The screenshot displays the Commvault web interface. On the left, the 'Manage' sidebar has 'Servers' selected, indicated by a red circle with the number 1. The main area shows a table of servers. The 'Actions' column for the 'mediaagent1' server is expanded, showing a context menu. The 'Check readiness' option is highlighted, indicated by a red circle with the number 3. Another red circle with the number 2 points to the 'Actions' icon in the table header.

Name ↑	Agent	Roles	Version	OS	Update status	Tags	Actions
esprod	Windows File System	Web Server, Command Center	11.32.8	Windows Server 2022 Standard	Up to date	No tags	⋮
mediaagent1	Windows File System	Storage MediaAgent	11.32.8	Windows Server 2022 Standard	Up to date	No tags	⋮

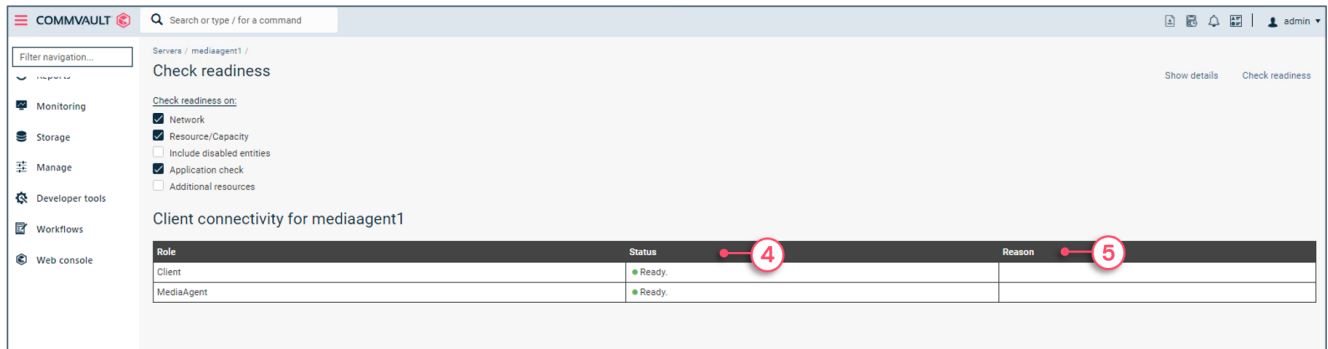
Context menu options for 'mediaagent1':

- Back up
- Restore
- View jobs
- Add software
- Upgrade software
- Retire
- Repair software
- Push network configuration
- Check readiness
- Restart services
- Send logs
- View logs
- Mark as non-infrastructure
- Manage tags

Connectivity to a server and storage data paths for configured subclients are managed at the server level. The Check Readiness command ensures that the CommServe server can communicate with the server. It also checks data path connectivity to all MediaAgents and paths for storage that are associated with subclients.

1. Select **Manage > Servers**
2. Click the **Actions** icon
3. Select **Check readiness**

Check Server Connectivity ...cont'd

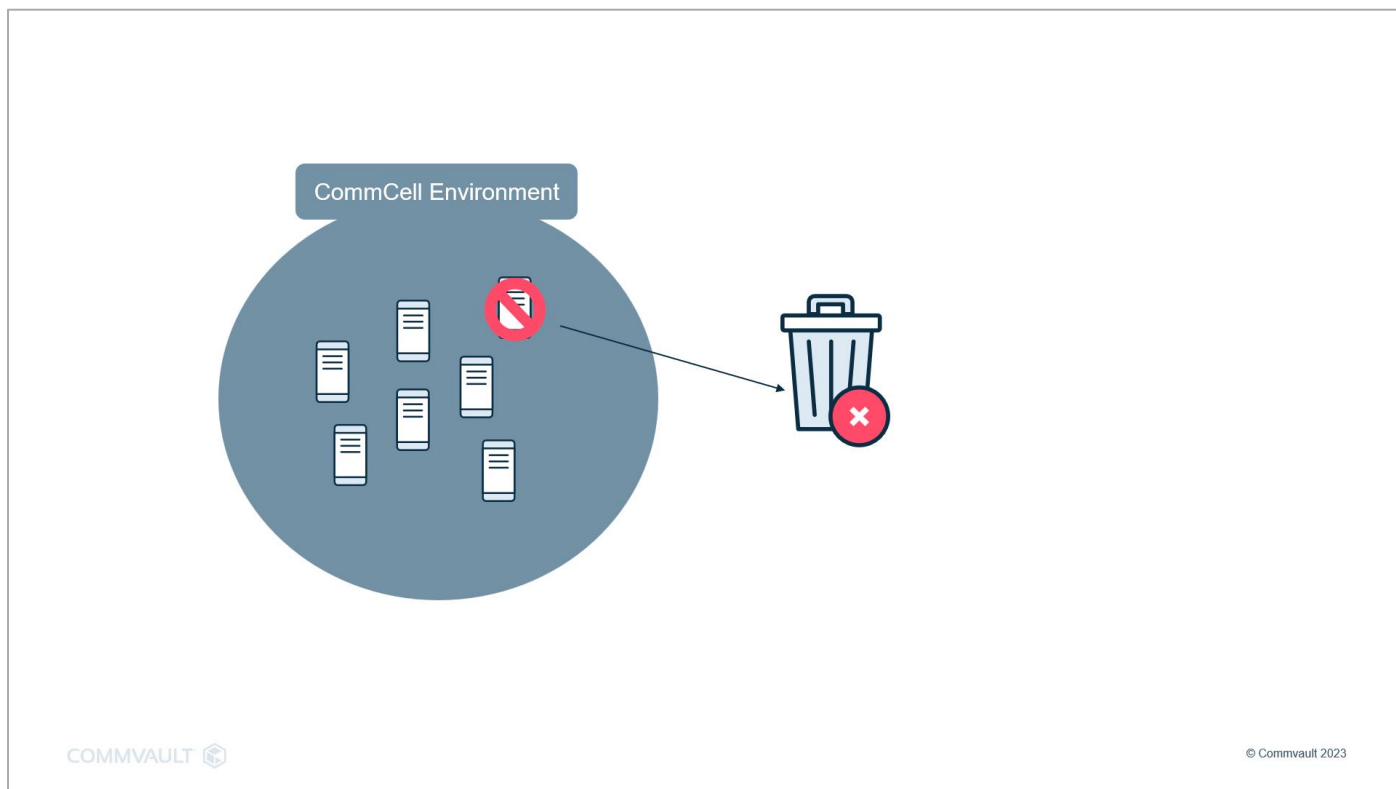


The screenshot shows the Commvault web console interface. The left sidebar contains navigation links: Monitoring, Storage, Manage, Developer tools, Workflows, and Web console. The main content area is titled 'Check readiness' and shows a table of client connectivity for 'mediaagent1'. The table has three columns: 'Role', 'Status', and 'Reason'. The 'Status' column is highlighted with a red circle and the number 4, and the 'Reason' column is highlighted with a red circle and the number 5. The table contains two rows: 'Client' and 'MediaAgent', both with a status of 'Ready'.

Role	Status	Reason
Client	Ready	
MediaAgent	Ready	

4. The result is displayed in the *Status column*
5. The *Reason column* describes reason for readiness failure

Retire a Server



Retiring a server removes the server completely from the CommCell environment. It is considered a destructive operation, since the data associated to that server is marked to be destroyed. During the next data aging operation, which is scheduled every day at noon by default, the server's data is physically purged from the disk or cloud storage. Data on tapes is also marked as aged, and tapes can be recalled as scratch media if empty.

All licenses for the server must be first released to allow the deletion. Consider releasing licenses instead of retiring a server if data needs to be preserved.

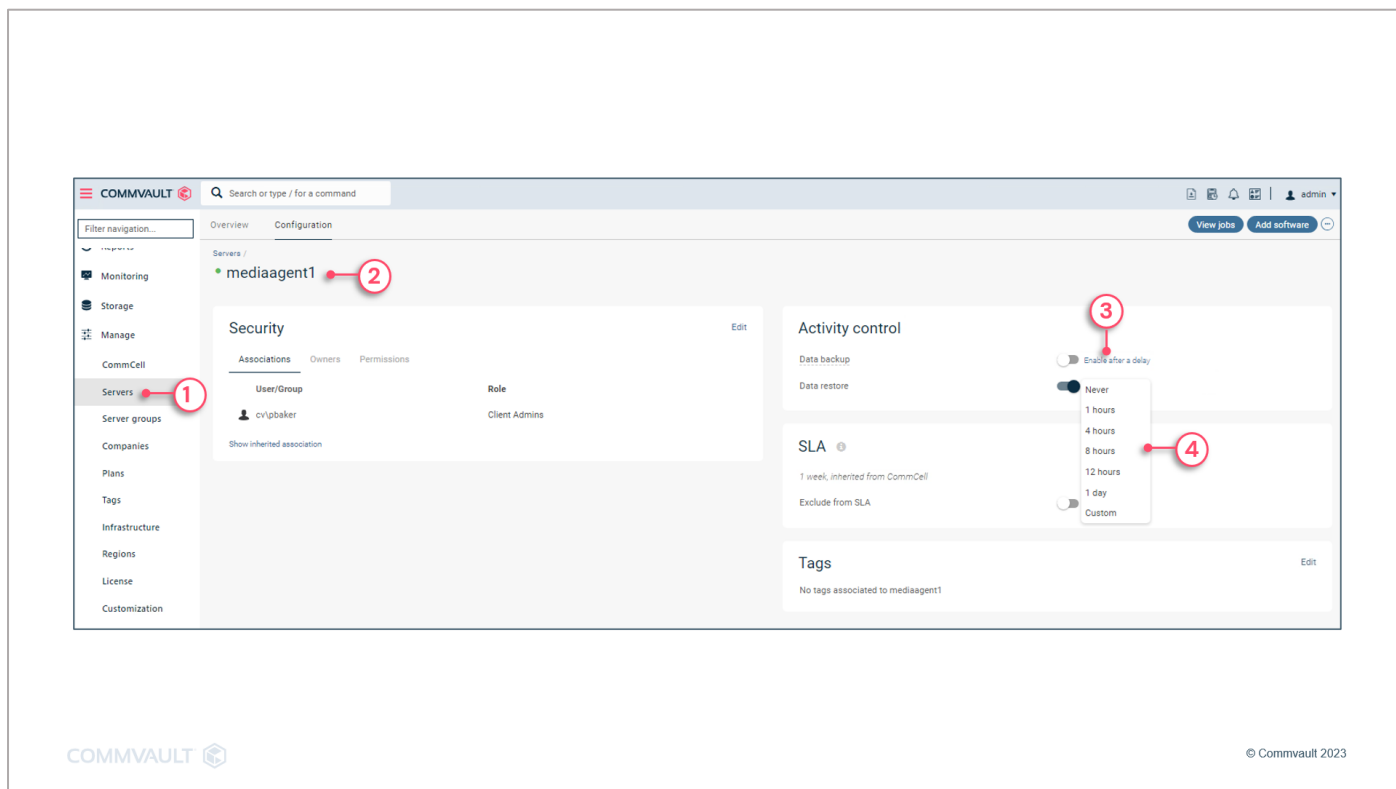
Retire a Server ...cont'd

The screenshot shows the Commvault web interface. On the left, the 'Manage' menu is expanded, and 'Servers' is selected, indicated by a red circle with the number 1. The main area displays a table of servers. The 'mediaagent1' server is selected, and the 'Actions' menu is open, with 'Retire' highlighted, indicated by a red circle with the number 2. A modal dialog box titled 'Retire mediaagent1' is displayed in the center. The dialog contains the following text: 'The retire action will release the license and uninstall the Commvault software from mediaagent1. If no backup data exists, then mediaagent1 will be deleted. Otherwise, mediaagent1 will be deconfigured but available for restores.' Below this text is a confirmation field with the prompt 'Type 'RETIRE' to confirm' and a red circle with the number 3. The field contains the text 'RETIRE'. At the bottom of the dialog are two buttons: 'CANCEL' and 'RETIRE'. The Commvault logo is visible in the bottom left corner of the interface.

Name	Agent	Roles	Version	OS	Update status	Tags	Actions
csprod	Windows File System	Web Server, Command Center	11.32.8	Windows Server 2022 Standard	Up to date	No tags	
mediaagent1	Windows File System	Storage MediaAgent	11.32.8	Windows Server 2022 Standard	Up to date	No tags	

1. Select **Manage > Servers**
2. Click the **Actions** icon and then click **Retire**
3. Type *Retire* and then click **RETIRE** to confirm

Server Activity Control



Data protection and data recovery jobs are enabled or disabled in the server properties view. If activity is disabled, an 'Enable after a Delay' button is displayed. This is used to automatically enable the activity on a specific date and time. Client activity control is useful when a client is offline since any scheduled operations are ignored if activity is disabled.

1. Select **Manage > Servers**
2. Select the server name
3. Toggle **Data backup** and **Data restore** to enable or disable activity
4. Toggle on **Enable after a delay** to re-enable activity after a specified time period, then choose the delay period

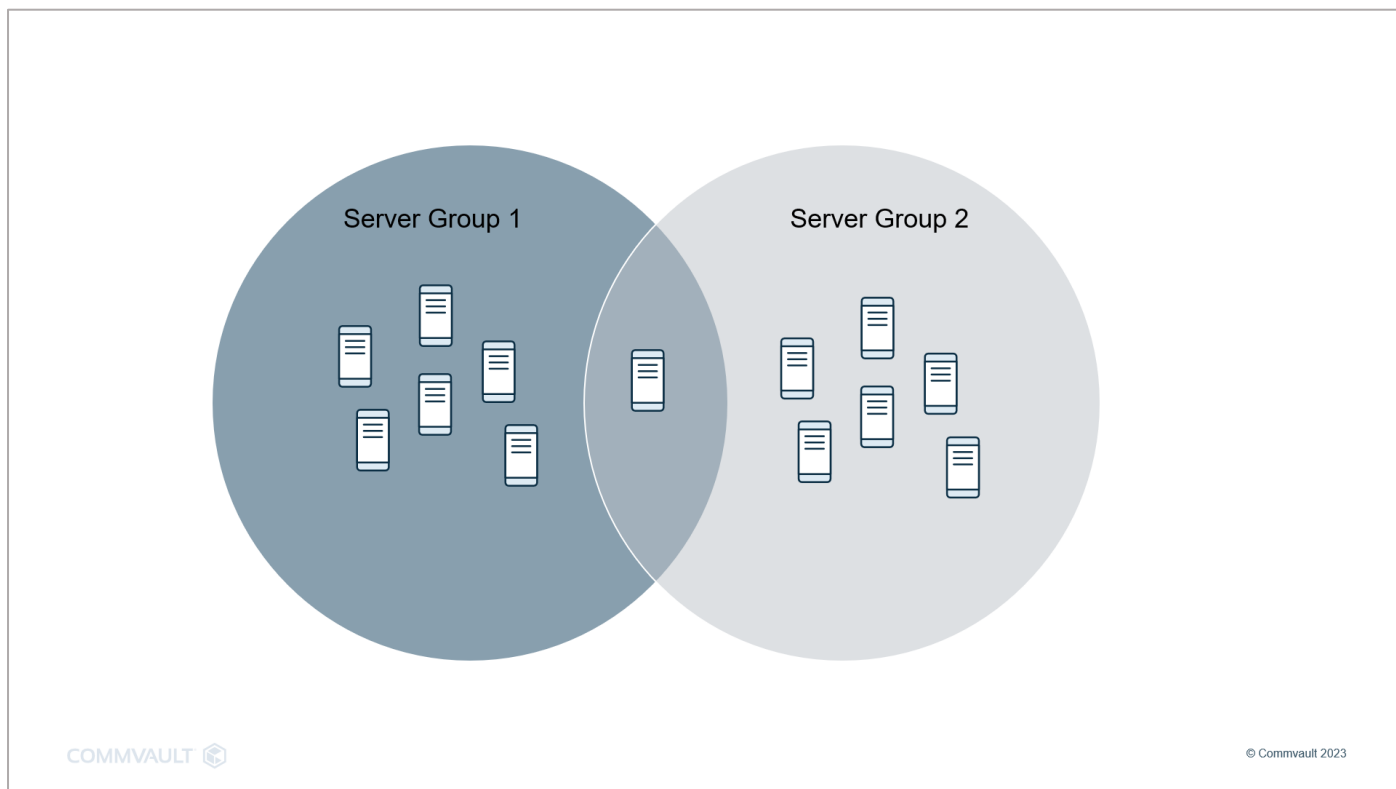
Server Update Status

Name ↑	Agent	Roles	Version	OS	Update status	Tags	Actions
☐ caprod	Windows File System, 1.3	Web Server, Command Center	11.32.8	Windows Server 2022 Standard	Up to date	No tags	⋮
☐ hyperv1	Virtual Server	MediaAgent	11.32.8	Windows Server 2022 Standard	Up to date	No tags	⋮
☐ mediaagent1	Windows File System	Storage MediaAgent, 1.4	11.32.8	Windows Server 2022 Standard	Up to date	No tags	⋮

Update statuses for servers are viewed from the Servers view of the Commvault Command Center. The current Commvault software version, service pack or release level, update status, and operating system platform are displayed for all clients providing summary information on their status.

1. Select **Manage > Servers**
2. Click **All**
3. The update status is displayed
4. Displays the software version of each server

Create a Server Group



Server groups add simplicity to CommCell administration. A server can be added to one or more server groups. This makes it easy to find servers, configure group security for multiple servers at once, and manage activity control for multiple servers.

There are two association methods for server groups. First, computers can be associated manually to a group. This method requires manual intervention to associate servers to the group. This is accomplished during the agent installation or by editing the server group.

Automatic association rules can also be used. This efficient method automatically associates servers to a group based on a set of predefined rules.

Create a Server Group ...cont'd

Filter navigation...

COMMVAULT Search or type / for a command

Server groups

2 Add server group

All

+ Add filter

Name ↑	Association	Tags	Actions
Automatic TPPM Clients	Manual	No tags	⋮
Infrastructure	Automatic	No tags	⋮
Laptop Clients	Automatic	No tags	⋮
Media Agents	Automatic	No tags	⋮
Proxy Clients	Automatic	No tags	⋮

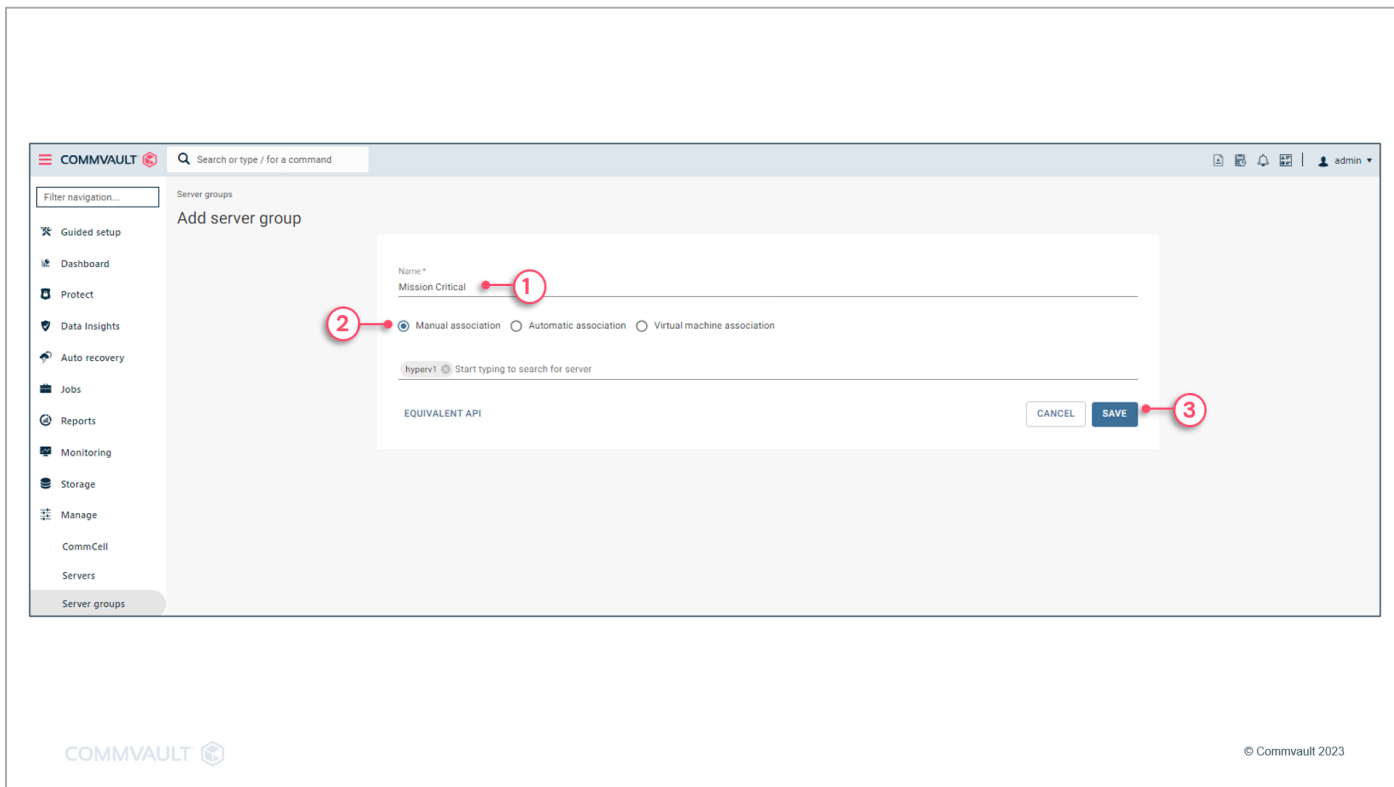
1

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1. Select **Manage > Server groups**; Existing server groups are displayed in this window
2. Click **Add server group** to add a new server group

Manual Association



1. Enter a name for the new group
2. Select **Manual association**
3. Click **SAVE** to create the server group

Automatic Association

The screenshot shows the 'Add server group' form in the Commvault interface. The form is titled 'Add server group' and is part of the 'Server groups' section. It includes a search bar at the top with the text 'Search or type / for a command'. The left sidebar contains a 'Filter navigation...' search bar and a list of navigation items: Guided setup, Dashboard, Protect, Data Insights, Auto recovery, Jobs, Reports, Monitoring, Storage, Manage, CommCell, Servers, and Server groups. The form itself has the following fields and options:

- Name ***: A text input field containing 'Mission Critical' (callout 1).
- Association type**: Three radio buttons: 'Manual association', 'Automatic association' (selected), and 'Virtual machine association' (callout 2).
- Client scope**: A dropdown menu with 'Clients in this CommCell' selected (callout 3).
- Rule groups**: A section with an 'Add rule group' button (callout 4) and a dropdown menu with 'all' selected (callout 5).
- Rule group**: A section with an 'Add rule' button and a dropdown menu with 'all' selected.
- Package installed any in**: A dropdown menu with '1 item(s) selected'.
- EQUIVALENT API**: A text input field.
- Buttons**: 'CANCEL', 'SAVE', and 'PREVIEW' buttons at the bottom right.

1. Enter a name for new server group
2. Select **Automatic association**
3. Select the desired client scope from dropdown list
4. Click **Add rule group** to add a rule
5. Select the criteria the rule will use; use scroll bar to view additional rule options

VM Association

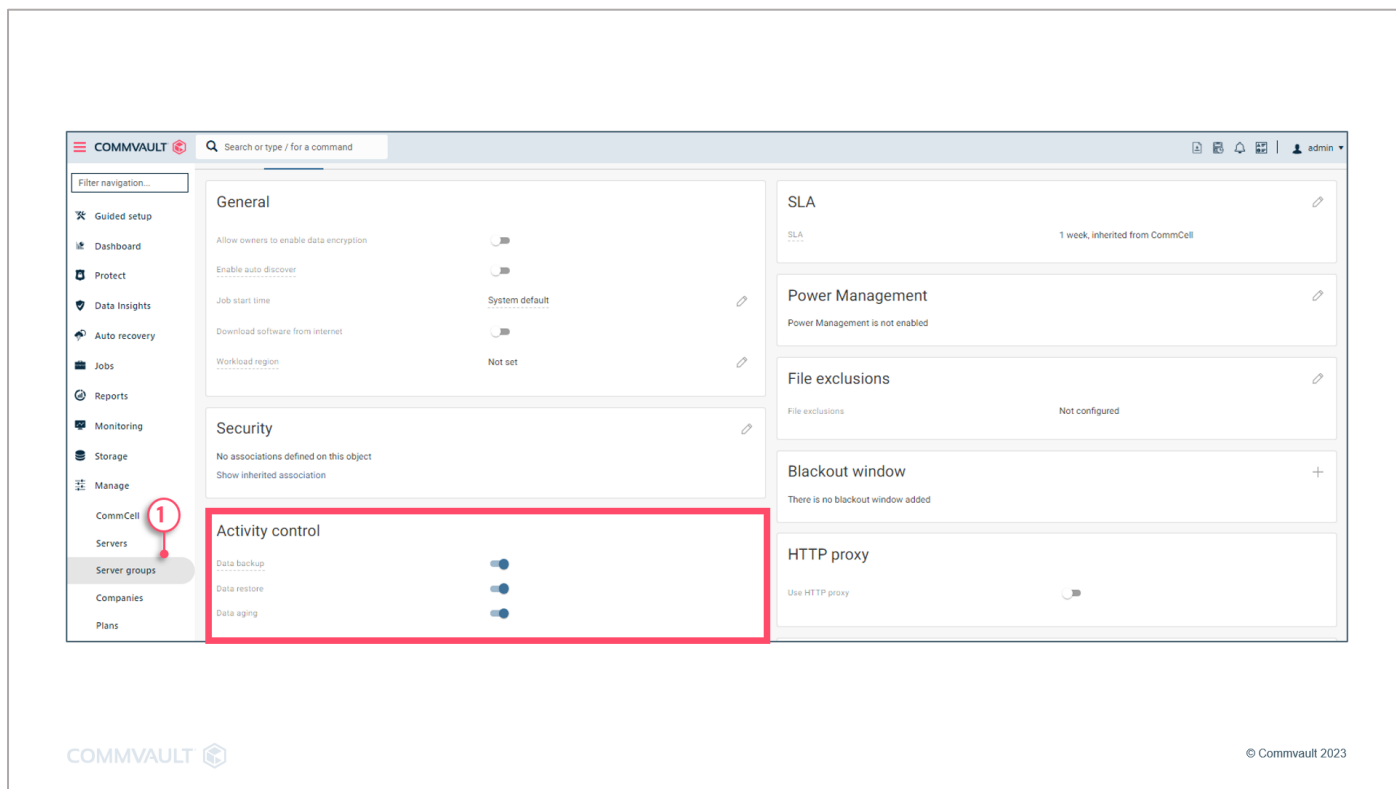
The screenshot shows the 'Add server group' form in the Commvault interface. The form is titled 'Add server group' and is part of the 'Server groups' section. It includes a search bar at the top and a sidebar with navigation options. The main form area contains the following fields and options:

- Name ***: Mission Critical
- Association Type**: ☐ Manual association, ☐ Automatic association, ☒ Virtual machine association
- Vendor ***: Microsoft Hyper-V (marked with a red circle and '1')
- Hypervisor ***: HyperV1 (marked with a red circle and '1')
- Match rule**: ☐ all, ☒ any
- Content**: A table with columns 'Type', 'Name', and 'Actions'. It contains one row: 'Virtual machine' (marked with a red circle and '2') with the name 'vmwinstq1'.

At the bottom of the form, there are buttons for 'EQUIVALENT API', 'PREVIEW', 'CANCEL', and 'SAVE'.

1. For *Virtual machine association*, select the vendor and hypervisor
2. Select the required virtual machines

Server Group Activity Control



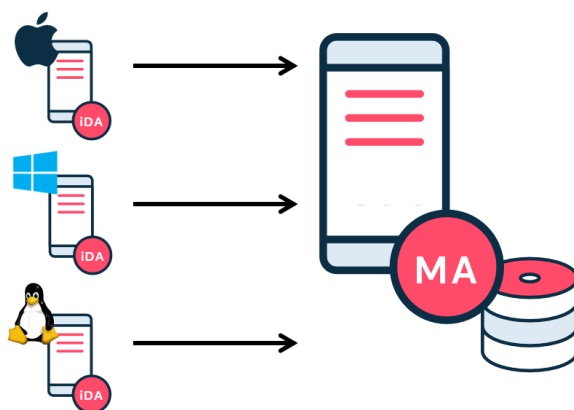
Activity can be controlled for an entire server group. This is useful when maintenance requires activity to be disabled and later re-enabled on several computers at once.

1. Select **Manage > Server groups** and click the server group name

In the *Activity control* section, use the toggles to disable the activities. Once disabled you are given the option to choose *Enable after a delay* to select the time period.

Files and Network Storage Protection

Operation System File Protection



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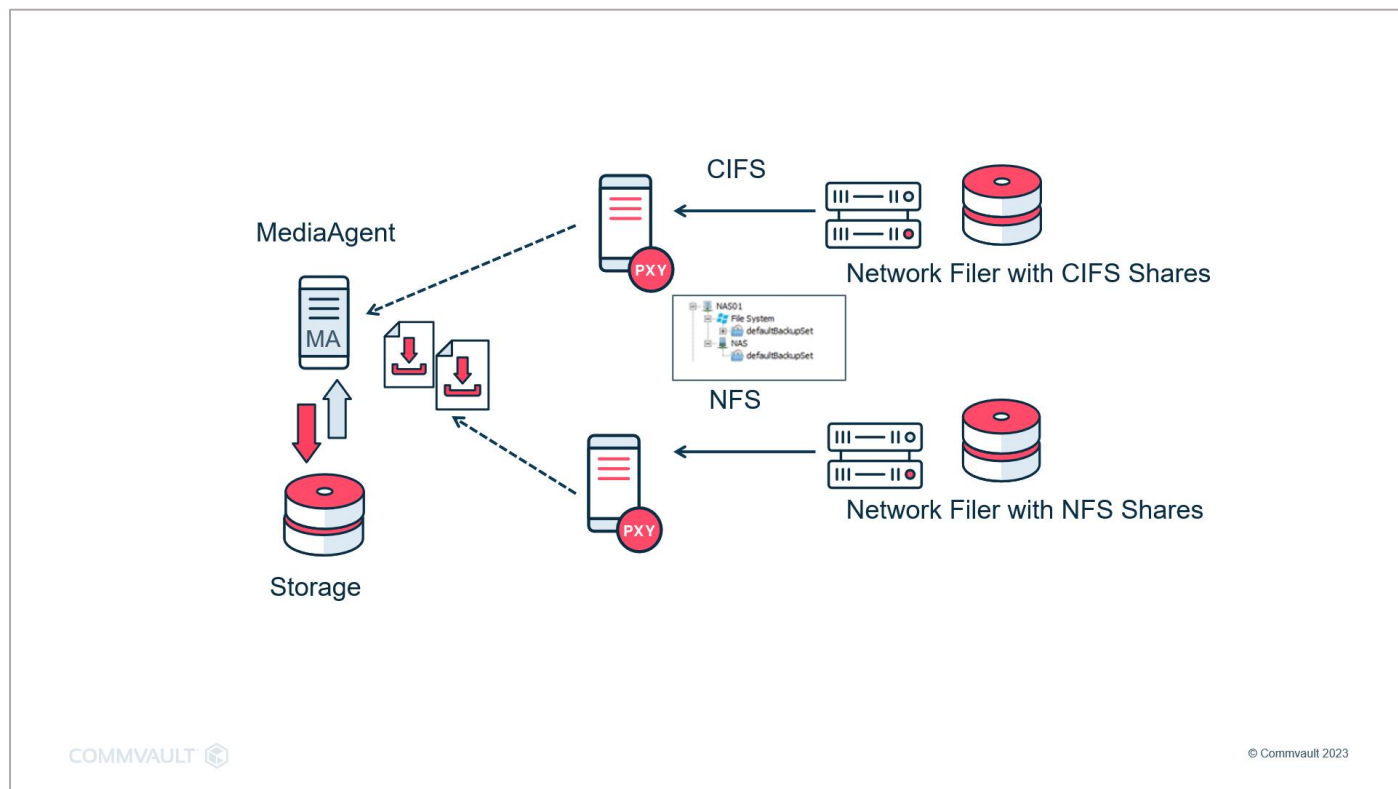
The file server solution provides tools to protect UNIX and Windows servers as well as network storage devices that share content via SMB or NFS file systems. Once the Commvault Command Center core setup is complete, the file server solution can be configured.

Commvault provides the following features so that you can efficiently manage your data protection:

- Job monitoring
- Alerts
- Events
- Resource reports
- File archiving and stubbing – Last access time
- Ransomware protection – Multi-level protection and alerting
- File system synchronization between two similar systems

During configuration, a backup plan can be created. Otherwise, the system uses the default plan. The plan defines protection settings such as the storage targets to receive data as well as the retention of the data.

Network File Protection

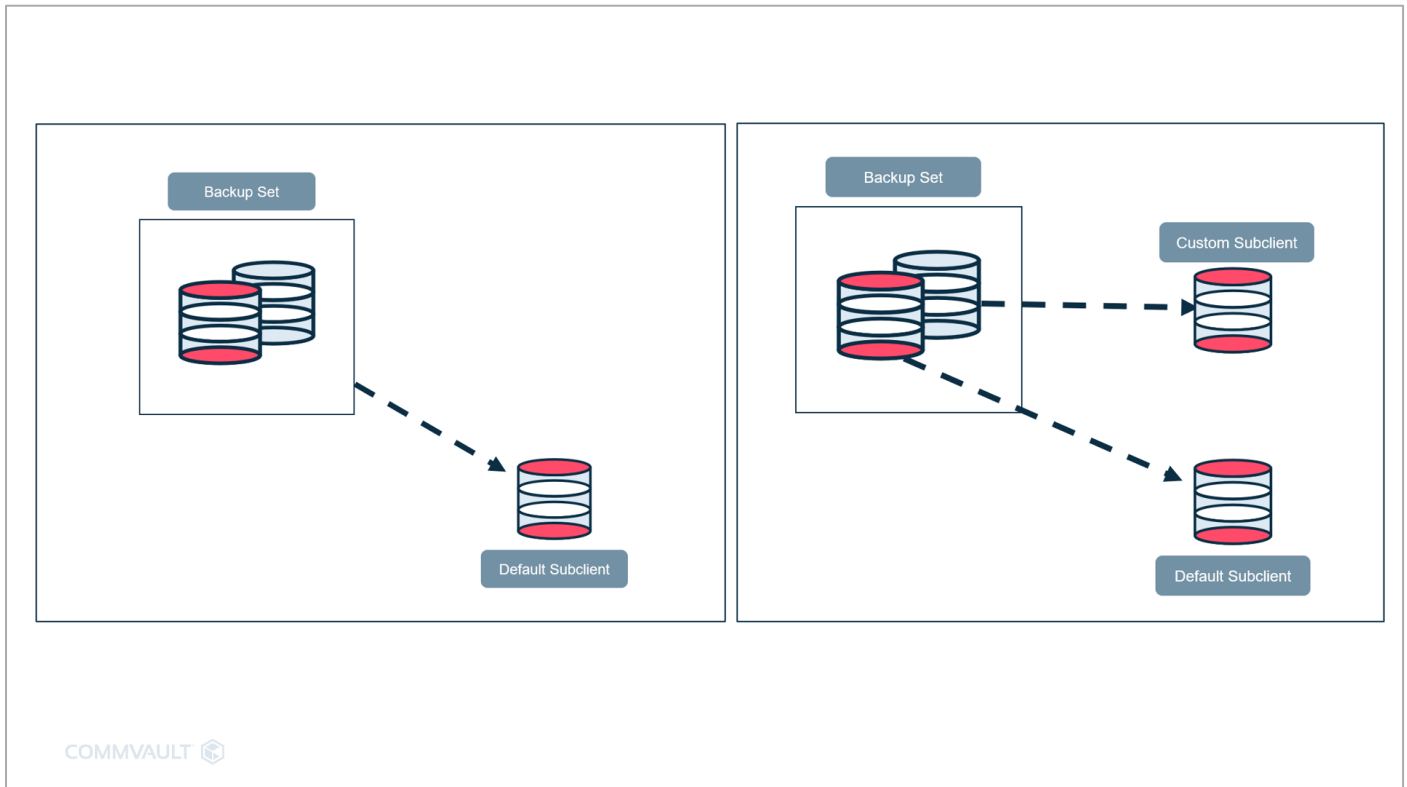


A MediaAgent can be used to protect network storage leveraging the hardware capabilities of the storage hardware. Support for Clustered File Systems, Big Data stores, and Metro-Cluster support are integrated into the Commvault solution through agent capabilities. Vendor options vary.

NFS/SMB mounts support:

- Snapshot
- NDMP backup protocol
- Block-level backups
- File System backups

Subclients



Subclients are used to define data that is protected in a container format. Each subclient container manages specific content within a backup set. Each backup set has one or more subclients.

Default Subclient

Most agents have a default subclient that acts as a catch all for all data managed within a backup set. This means the default subclient automatically detects and protects all data the agent is responsible to protect. When custom subclients are defined, any data managed by the custom subclients is automatically excluded from the default subclient. This is the concept of mutual exclusivity of content within a backup set, such that data cannot be defined in multiple subclients within the backup set. Best practice is to protect the data to one storage target and use secondary copies to create additional copies of data. There are situations where protecting data from the source location multiple times may be required. To accomplish this, you can create additional backup sets.

Define Contents for the Subclient

The contents of a default subclient, for most backup agents, is a \ (Windows-based) or / (Linux/UNIX based). This represents an auto-detection functionality that protects any newly added volumes. If the default subclient's content is modified, the \ or / is removed and auto-detection is disabled. Do not modify the contents of the default subclient. Use the filter function to exclude unwanted drives.

Add content by using *Browse* or by manually entering it using *Add/Edit Paths*.

Use *Add Paths* to enter UNC paths to protect data on systems that do not have Commvault agents installed. Then use *As user* to enter a user account with proper permissions to read the data from the shared location. This feature is only recommended when protecting small amounts of data.

The option to 'Backup System State' is used to protect system state data for Windows servers. By default, the default subclient automatically protects system state data. If required, a separate subclient can be defined to specifically protect system state. Only one subclient within a backup can be designated to protect system state data.

Define Content Using Wildcards and Patterns

When defining content using *Add Paths*, regular expressions (or wildcard characters) can be used. Wildcard expressions are characters such as asterisk (*) or question mark (?). Regular expressions include patterns such as [a-f] or *.[l-n]df. The terms are interchangeable. Use the regular expressions for defining content at any level in the data path. For example:

- F:\Users
- [A-L]* or *.pst.

The expression ** represents any folder structure. For instance, C:**\data\ means that any data, from the C: drive, in any folder structure, is backed up.

Edit Subclient Content

The screenshot displays the Commvault console interface. The left sidebar shows the navigation menu with 'Protect > File servers' highlighted. The main area shows a table of file servers. The first table lists file servers with columns: Name, Configured, Last backup, Application size, Plan, SLA status, Tags, and Actions. The second table shows the configuration for a selected file server, including status, client readiness, last successful backup, oldest backup time, application size, savings percentage, company, and workload region. The third table lists subclients with columns: Name, Backup content, Excluded from backup, Last backup, Last backup size, Plan, Status, and Actions. Red circles and numbers 1, 2, and 3 indicate the steps to edit subclient content.

1. Select **Protect > File servers**

2. Click the file server name

3. Click the subclient name

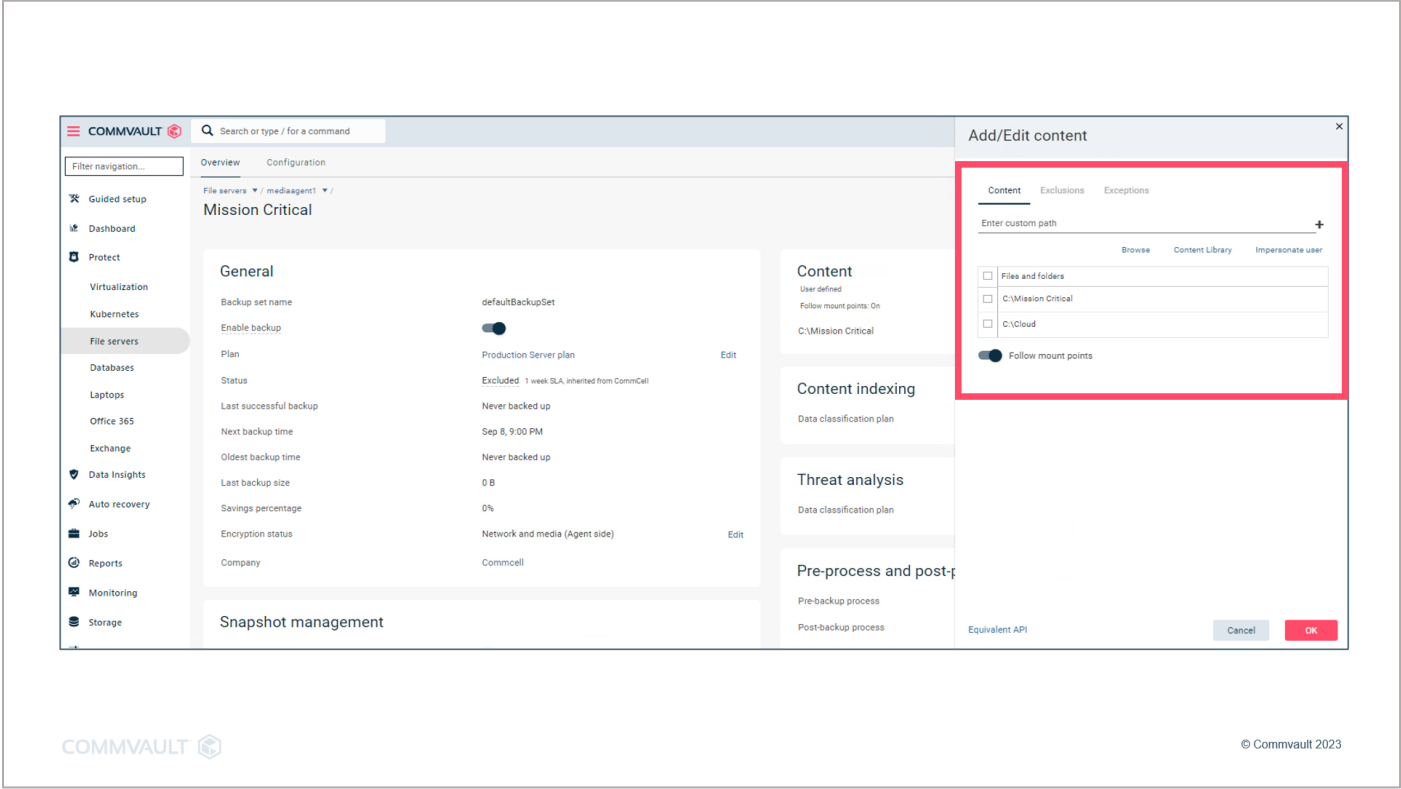
1. Select **Protect > File servers**
2. Click the file server name
3. Click the subclient name

Edit Subclient Content ...cont'd

The screenshot displays the Commvault console interface. On the left, the 'File servers' section is expanded, showing the 'Mission Critical' subclient. The 'Content' section is highlighted with a red circle 4. An 'Add/Edit content' dialog is open, showing the 'Content' tab with a 'Browse' button highlighted by a red circle 5. A 'Select a path' dialog is also open, showing a file tree with 'Mission Critical' highlighted by a red circle 6. The 'Follow mount points' checkbox is checked in the 'Add/Edit content' dialog.

4. In the *Content* section, click **Edit**
5. Click **Browse** to navigate to the location
6. Select the required content

Edit Subclient Content ...cont'd



This area displays the added content.

Add Subclients

The screenshot shows the Commvault console interface. On the left, the 'File servers' menu item is highlighted with a red circle labeled '1'. The main area displays the 'Configuration' tab for a file server, showing details like Status, Client readiness, Last successful backup, Oldest backup time, Application size, and Savings percentage. Below this, the 'Subclients' section is visible, containing a table of subclients. The table has columns for Name, Backup content, Excluded from backup, Last backup, Last backup size, Plan, Status, and Actions. The 'default' subclient is expanded, showing its details. A red circle labeled '2' points to the 'Subclients' table. In the top right corner, the 'Add subclient' button is highlighted with a red circle labeled '3'.

Name	Backup content	Excluded from backup	Last backup	Last backup size	Plan	Status	Actions
default	\ - All contents, [System state]	No exclude content	Never backed up	0 B	Production Server plan	Excluded	
Plan							
Production Server plan							
Last successful backup		Never backed up					
Last backup size		0 B					
Mission Critical							
C:\Mission Critical		No exclude content	Never backed up	0 B	Production Server plan	Excluded	

When new subclients are added, the content defined in the new subclient is automatically removed from the default subclient. Content within a backup set cannot be defined in more than one subclient, except for VSA subclients.

1. Select **Protect > File servers** and click the server name for which you want to create a new subclient
2. Subclients are listed in this window
3. Click **Add subclient** to add new subclient

Add Subclients ...cont'd

The screenshot displays the Commvault console interface. On the left is a navigation sidebar with categories like Guided setup, Dashboard, Protect, Virtualization, Kubernetes, File servers, Databases, Laptops, Office 365, Exchange, Data Insights, Auto recovery, Jobs, Reports, Monitoring, and Storage. The main area shows the 'Overview' tab for a client, with fields for Status, Client readiness, Last successful backup, Oldest backup time, Application size, Savings percentage, Company, and Workload region. Below this is a 'Subclients' table.

Name	Backup content	Excluded from backup	Last backup	Last backup size
default	\ - All contents, [System state]	No exclude content	Never backed up	0 B
Plan	Production Server plan			
Last successful backup	Never backed up			
Last backup size	0 B			
Mission Critical	C:\Mission Critical, C:\Cloud	No exclude content	Never backed up	0 B

On the right, the 'Add subclient' dialog box is open. It contains the following fields and controls, each marked with a red circle and a number:

- 4: Name field (containing 'Other Data')
- 5: Backup set dropdown (containing 'defaultBackupSet')
- 6: Plus sign (+) to add custom paths
- 7: Browse button
- 8: Exclusions tab/button

Below these fields are checkboxes for 'Files and folders' and 'C:\Cloud'. At the bottom, there are 'Pre-backup process' and 'Post-backup process' input fields, an 'Impersonate user' toggle, and 'Equivalent API' checkbox. 'Cancel' and 'OK' buttons are at the bottom right.

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4. Enter a name
5. Choose a protection plan
6. Enter custom path locations and click + to add them to the list
7. Click **Browse** to browse and select content
8. Click **Exclusions** to define data to exclude

Add Subclients ...cont'd

The screenshot shows the Commvault console interface. On the left is a navigation menu with options like Guided setup, Dashboard, Protect, Virtualization, Kubernetes, File servers, Databases, Laptops, Office 365, Exchange, Data Insights, Auto recovery, Jobs, Reports, Monitoring, and Storage. The main area displays the 'Subclients' configuration. A table lists subclients with columns: Name, Backup content, Excluded from backup, Last backup, and Last backup size. The 'Add subclient' dialog box is open on the right, with tabs for Content, Exclusions, and Exceptions. The 'Exceptions' tab is selected, showing a field for 'Enter custom path' (marked with a red circle 9) and a section for 'Files and folders' (marked with a red circle 10). At the bottom right of the dialog, the 'OK' button is highlighted with a red circle 11.

Name	Backup content	Excluded from backup	Last backup	Last backup size
default	\- All contents, [System state]	No exclude content	Never backed up	0 B
Plan: Production Server plan				
Last successful backup: Never backed up				
Last backup size: 0 B				
Mission Critical	C:\Mission Critical, C:\Cloud	No exclude content	Never backed up	0 B

Subclient filters can be defined for Windows and UNIX subclients to allow for the exclusion and exception of specific folders and objects that do not require protection. Wildcards and patterns are used to define exclusion filters and exceptions. Exclusion filters determines which folders and/or objects will be excluded from the subclient. Exceptions override exclusions. Any data defined in the exception are protected by the subclient. An example would be filtering a temporary directory by defining it in the exclusions section but adding an important folder to the exceptions section for backup.

9. Click **Exclusions** to define data to exclude
10. Exceptions can be applied to exclusions, providing granularity if needed
11. Click **OK** to create the subclient

Delete a Subclient

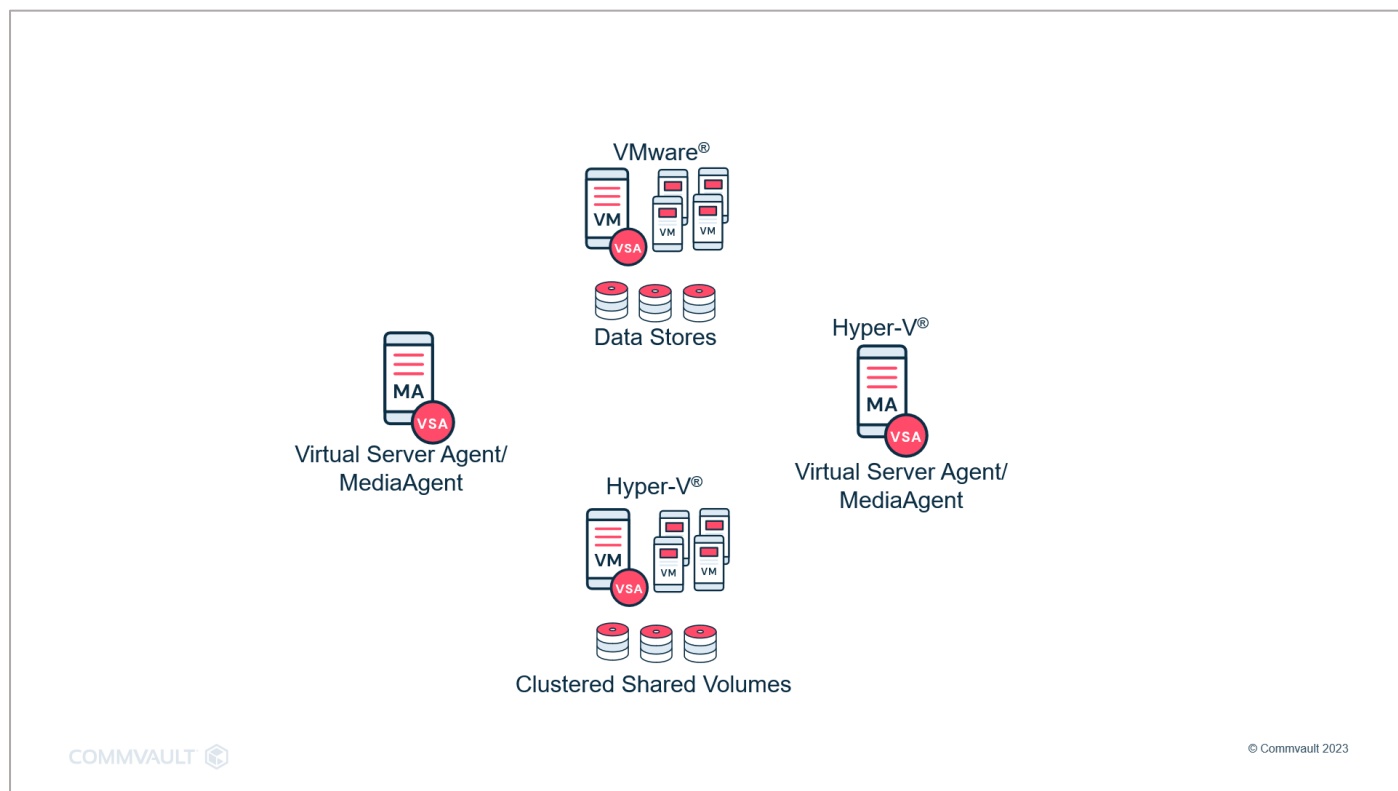
The screenshot shows the Commvault console interface. On the left sidebar, the 'File servers' menu item is highlighted with a red circle and the number 1. The main content area displays the 'Subclients' table. The table has columns: Name, Backup content, Excluded from backup, Last backup, Last backup size, Plan, Status, and Actions. A subclient named 'default' is listed. The 'Actions' column for this subclient has a dropdown menu open, showing options: 'Back up', 'Backup history', and 'Delete'. The 'Delete' option is highlighted with a red circle and the number 2.

Name	Backup content	Excluded from backup	Last backup	Last backup size	Plan	Status	Actions
default	\ - All contents, [System state]	No exclude content	Never backed up	0 B	Production Server plan	Excluded	Back up Backup history Delete

1. Select **Protect > File servers** and click the server name for which you want to delete a subclient
2. Click the **Actions** icon for the subclient you want to delete

Virtual Machines Protection

Virtualization Overview



Virtualization has become the standard of data center consolidation whether on-premises or in the cloud. As the number of virtual machines and the physical hosts they run on grows, a comprehensive protection strategy is required. Commvault software provides several protection methods for virtual environments on-premises and in the cloud. These methods provide a comprehensive enterprise hybrid protection strategy.

Virtual Server Agent (VSA)

The Commvault Virtual Server Agent (VSA) interacts with the hosting hypervisor to provide protection for the virtual machine. Agents do not need to be installed directly on the virtual machines, although installing restore-only agents provides a simplified method for restoring data back to the VM.

Depending on the hypervisor application being used and the virtual machine's operating system, different features and capabilities are available. The VSA interfaces with the hypervisor's APIs and provides capabilities inherent to the application.

Agent-Based Protection

Agent-based protection uses Commvault agents installed directly in the virtual machine. When an agent is installed in the VM, it appears as a regular client and the functionality is the same as an agent installed on a physical host.

The main advantage with this configuration is that all the features available with Commvault agents are used to protect data on the VM. For applications, using a Commvault agent provides complete application awareness of all data protection operations including streaming log backups, granular item-based protection, archiving and content indexing.

VSA Application-Aware Protection

VSA application-aware backups insert an application plugin into the VM during a VSA backup. When a VM backup runs, the plugin quiesces the application using a VSS snapshot. The VSA coordinator then communicates with the hypervisor to conduct a VM snapshot. This protection method provides a hybrid approach using the VSA to conduct data protection jobs, and agent-based functionality for recovery, like installing an agent directly in the VM.

The method to use depends on the virtual infrastructure, type of virtual machines being protected, and the data contained within the virtual machines. In most cases using the Virtual Server Agent (VSA) is the preferred protection method. For specific virtual machines, using application-aware backups or an agent directly installed within the VMs is recommended. For mission critical virtual machines, large virtual machines, or virtual machines with high I/O processes, the IntelliSnap feature is used to coordinate hypervisor software snapshots with array hardware snapshots to efficiently protect virtual machines while minimizing the performance impact of the virtual infrastructure.

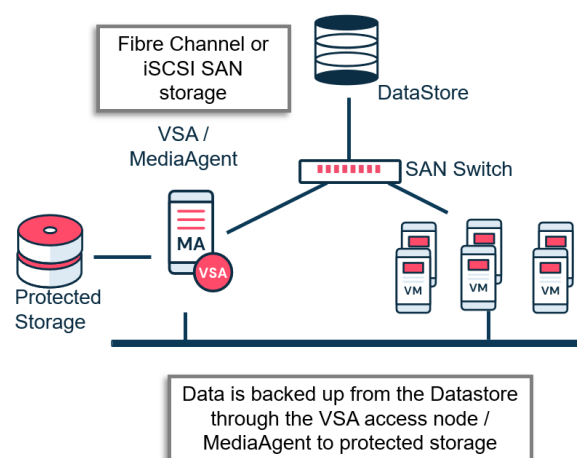
VMware Transport Modes

The VMware VADP framework provides several transport modes to protect virtual machines.

- SAN transport mode
- NAS transport mode
- HotAdd mode
- NBD and NBD SSL mode

Each of these modes has their advantages and disadvantages. Variables such as physical architecture, source data location, ESX resources, network resources and VSA proximity to MediaAgents and storage influence which mode is best to use. It is also recommended to consult with Commvault for design guidance when deploying Commvault software in a VMware environment.

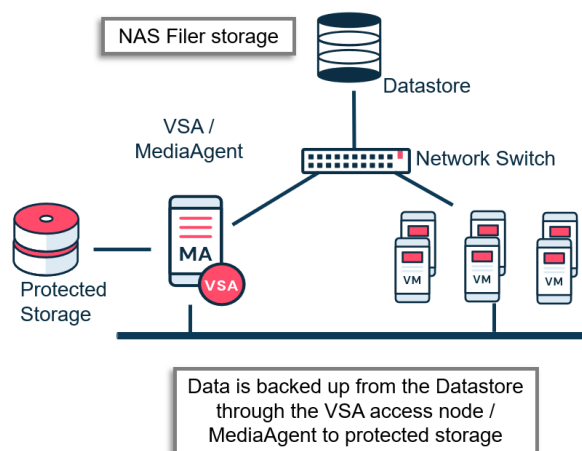
SAN Transport Mode



SAN transport mode is used on a VSA access node with direct Fibre Channel or iSCSI access to snapshot VMs in the source storage location. This mode provides the advantage of avoiding network movement of VM data and eliminates load on production ESX servers.

Virtual machines are backed up through the VSA and to the MediaAgent. LAN-free backups can be performed if the VSA is installed on an access node configured as a MediaAgent with direct access to storage. For best performance the VSA needs a dedicated HBA to access the VMDK files. If an iSCSI SAN is used, use a dedicated network interface card on the VSA for access to the SAN.

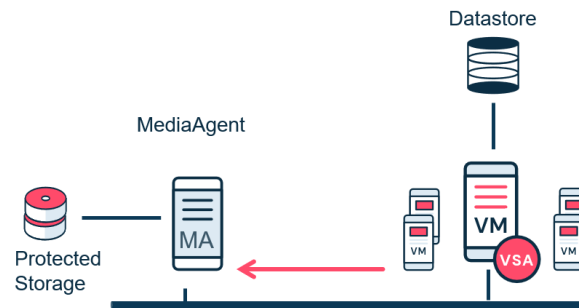
NAS Transport Mode



NAS Transport Mode is used on a VSA access node with direct network access to the datastore NFS volume exports in the source NAS filer. This mode provides the advantage of avoiding data movement of VM data through the production ESX servers.

Virtual machines are backed up through the VSA and to the MediaAgent. The VSA access node is typically installed directly on the physical MediaAgent. For best performance the VSA needs a dedicated network interface card for access to the NAS.

HotAdd Mode

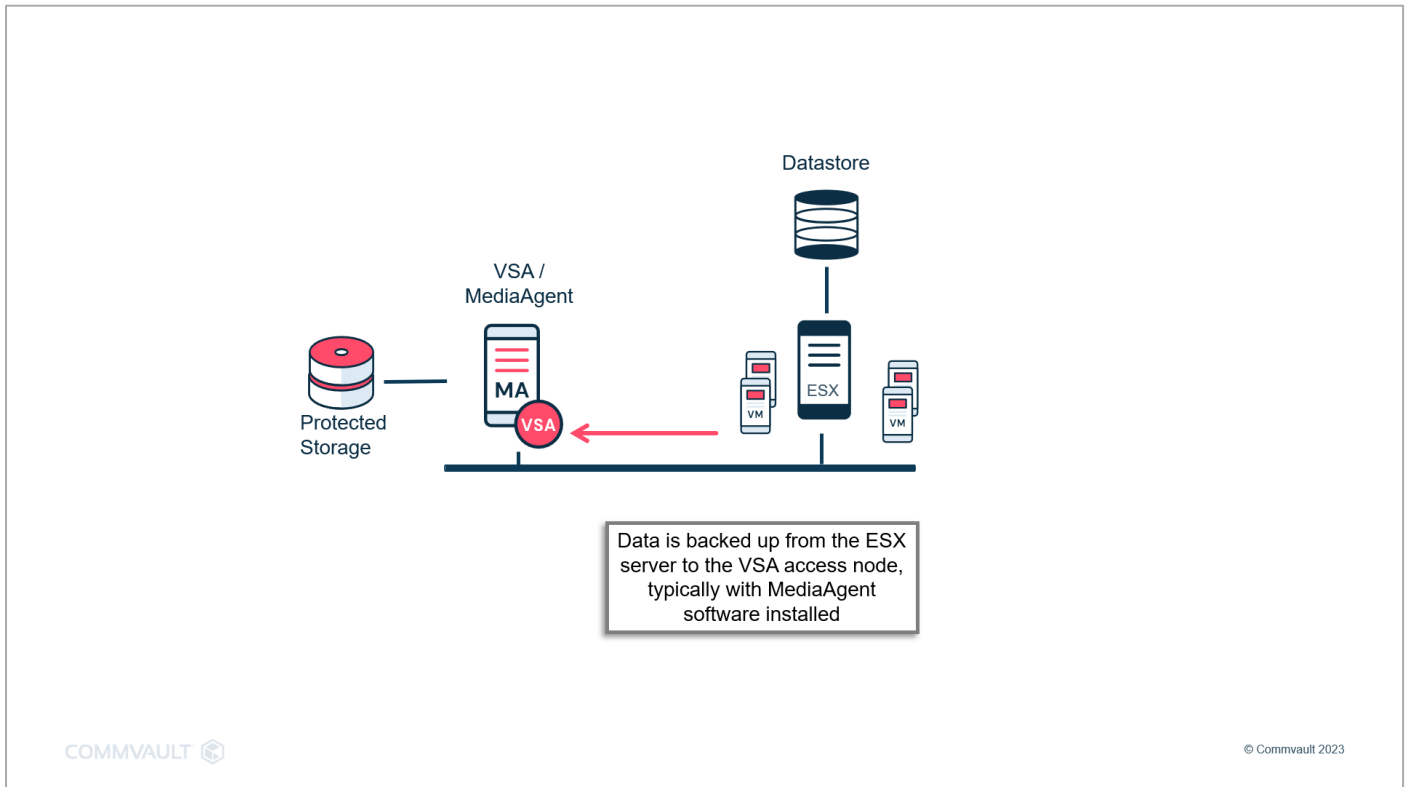


HotAdd mode uses a virtual VSA in the VMware environment. This requires all data to be processed and moved through the VSA access node on the ESX server. HotAdd mode has the advantage of not requiring a physical VSA access node and does not require direct SAN access to storage. It works by adding virtual disks to the VSA access node and backing up the disks and configuration files to protected storage.

A common method of using HotAdd mode is to use Commvault deduplication with client-side deduplication, DASH Full, and incremental forever protection strategy. Using Change Block Tracking (CBT), only changed blocks within the virtual disk have signatures generated and only unique block data are protected.

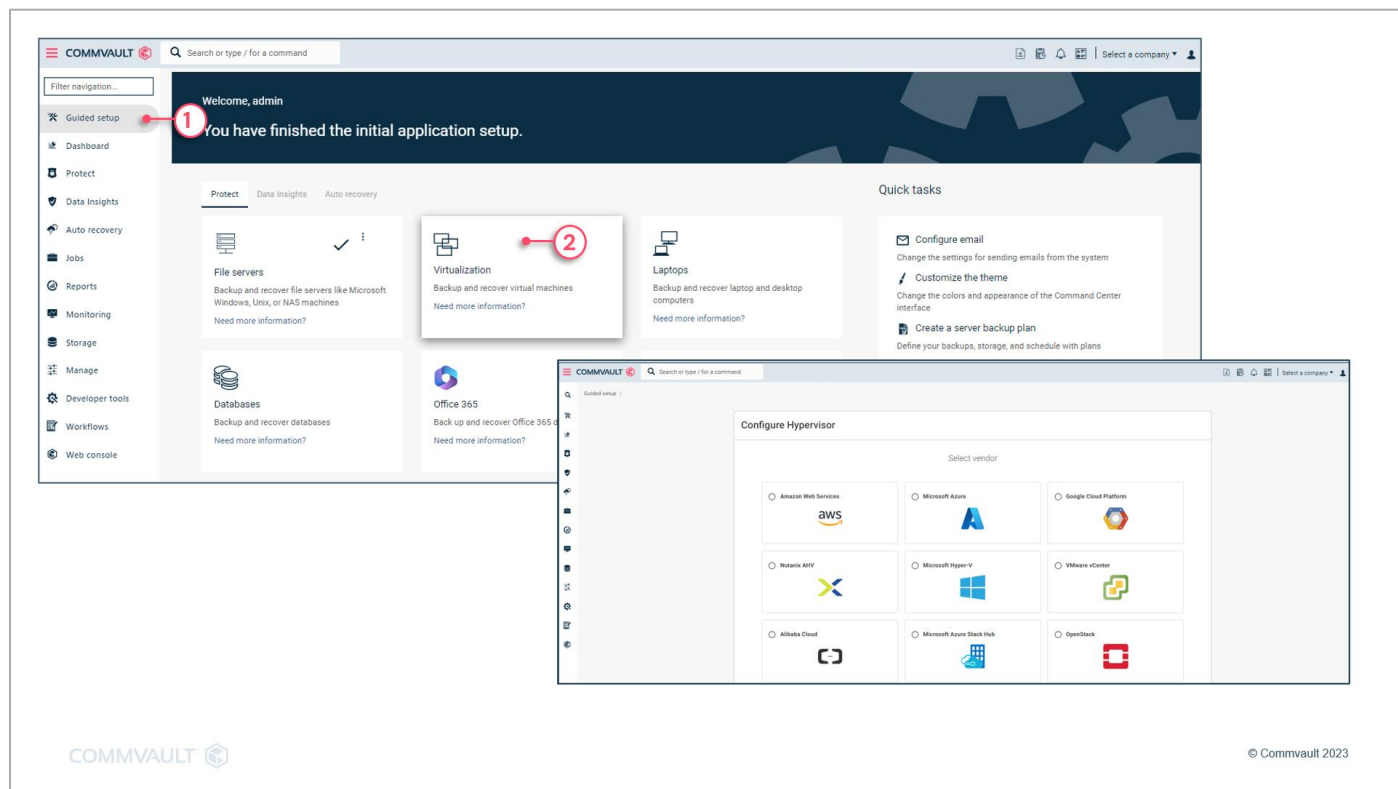
This mode is also useful when there is no physical connectivity between the physical VSA access node and the datastore storage preventing the use of SAN transport mode. Some examples of such scenarios are when using NFS datastores or using ESX host's local disk storage to host datastores.

NBD Mode



NBD mode is the simplest method to protect virtual machines. NBD mode uses a VSA access node installed on a physical host. The VSA connects to VMware and snapshots will be moved from the ESX server over the network to the VSA access node. This method requires adequate network resources.

Add the Virtualization Feature



Add a Hypervisor

Once the VSA software has been installed on all the access nodes, the hypervisor (also known as an instance) can be configured. One or many hypervisors from any supported vendor can be added in a CommCell environment.

Add the Virtualization Feature

1. Select **Guided setup**
2. Click the **Virtualization** tile

Select the vendor.

Add the Virtualization Feature ...cont'd

The screenshot shows the 'Configure Microsoft Hyper-V Hypervisor' wizard in the Commvault interface. The left sidebar contains a 'Filter navigation...' search bar and a list of navigation items: Guided setup, Dashboard, Protect, Data Insights, Auto recovery, Jobs, Reports, Monitoring, Storage, Manage, Developer tools, Workflows, and Web console. The main content area is titled 'Configure Microsoft Hyper-V Hypervisor' and features a progress indicator on the left with three steps: 1. Add Hypervisor (selected), 2. Add VM Group, and 3. Summary. The 'Add Hypervisor' step is further detailed with the following fields:

- Host name ***: Labeled with a red circle 3, containing the text 'Hyperv1.cv.lab'.
- Hypervisor display name ***: Labeled with a red circle 4, containing the text 'Hyperv1'.
- Credential ***: Labeled with a red circle 5, containing the text 'HyperV Access'.
- Access nodes ***: Labeled with a red circle 6, containing the text 'hyperv1'.

At the bottom of the form, there is a 'CANCEL' button and a 'DISCOVER NODES' button, which is highlighted with a red circle 7. The text 'EQUIVALENT API' is visible below the 'Access nodes' field.

3. Enter the full name of the vCenter server or host
4. Enter the client name to display in the console
5. Enter credentials with required permissions
6. Choose the VSA access node
7. Click **DISCOVER NODES**

Add the Virtualization Feature ...cont'd

COMMVAULT Search or type / for a command

Filter navigation... Guided setup /

Guided setup

- Dashboard
- Protect
- Data Insights
- Auto recovery
- Jobs
- Reports
- Monitoring
- Storage
- Manage
- Developer tools
- Workflows
- Web console

Configure Microsoft Hyper-V Hypervisor

1 Add Hypervisor

2 Add VM Group

3 Summary

Add VM Group

Name* 8

Production VMs

Content Add

Type	Name	Actions
☐	Virtual machine	vmwingsql1

Plan* 10

Production Server plan

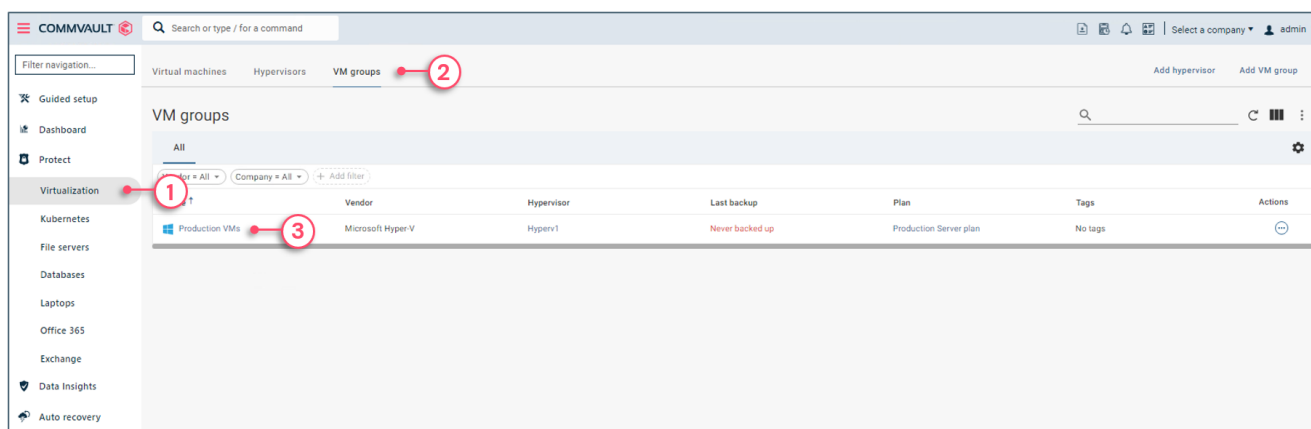
EQUIVALENT API

PREVIEW

CANCEL PREVIOUS NEXT 11

8. Enter a name for the VM group
9. Check the boxes to choose which virtual machines to protect
10. Choose the backup plan
11. Click **NEXT** to complete

Subclient Filters for VM



Subclient filters are used to filter virtual machines or virtual machine disks. This provides the necessary granularity when using auto-discovery rules. For example, a subclient can be defined to discover all VMs in a cluster, for which VMs not requiring protection can be filtered out.

1. Select **Protect > Virtualization**
2. Click the **VM groups** tab
3. Click the subclient name

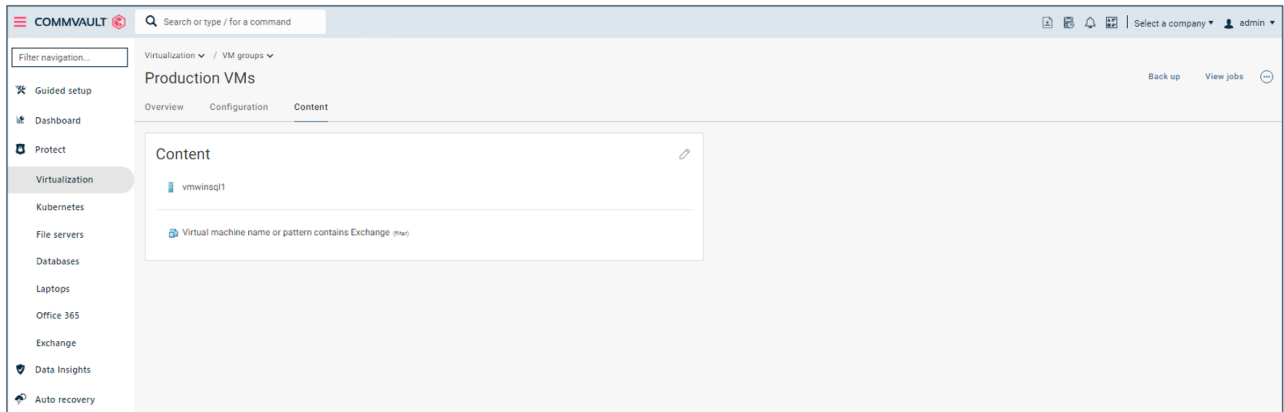
Subclient Filters for VM ...cont'd

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4. In the *Content* section click the **Edit icon**
5. Toggle on **Define filters**
6. Click **Add virtual machines** to manually select VMs to filter
7. Choose the browsing mode
8. Check the boxes to choose the VMs to filter

Subclient Filters for VM ...cont'd



Defined filters are displayed in the *Content* section.

View VM Group Job History

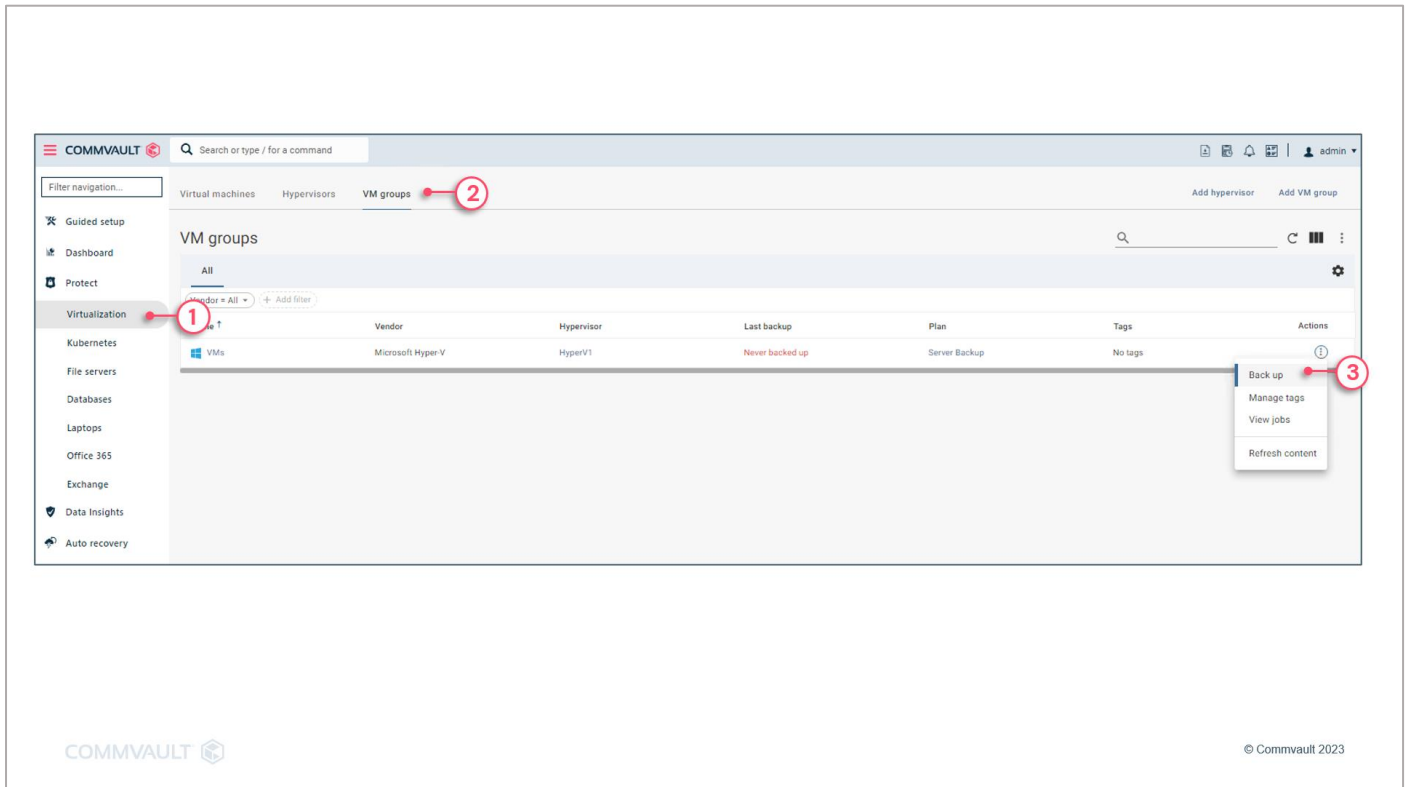
The screenshot displays the Commvault web interface. In the top-left sidebar, the 'Protect' section is expanded, and 'Virtualization' is selected. The main content area shows the 'VM groups' tab. A table lists VM groups, with one entry highlighted. The 'Actions' column for this entry is expanded, showing a dropdown menu with 'View jobs' selected. Below this, the 'Job history' view is shown, featuring a table of job details.

Job id	Operation	Server	Agent type	Subclient	Size	End	Elapsed	Status	Error description	Actions
37	VM Admin Job(Back...	HyperV1_1	Virtual Server	VMs	21.73 GB	Sep 10, 2023, 9:32:5...	3 minutes 45 seconds	Completed		

1. Select **Protect > Virtualization**
2. Click the **VM Groups** tab
3. Click the **Actions** icon and then click **View jobs**

The view displays the history of backups, restores, and Live Sync replications.

Backup from a VM Group



Virtual Machine Backups

There are two places from which manual VM backups can be initiated in the Commvault Command Center. It is important to understand the difference between the two as it may launch a backup for more VMs than required.

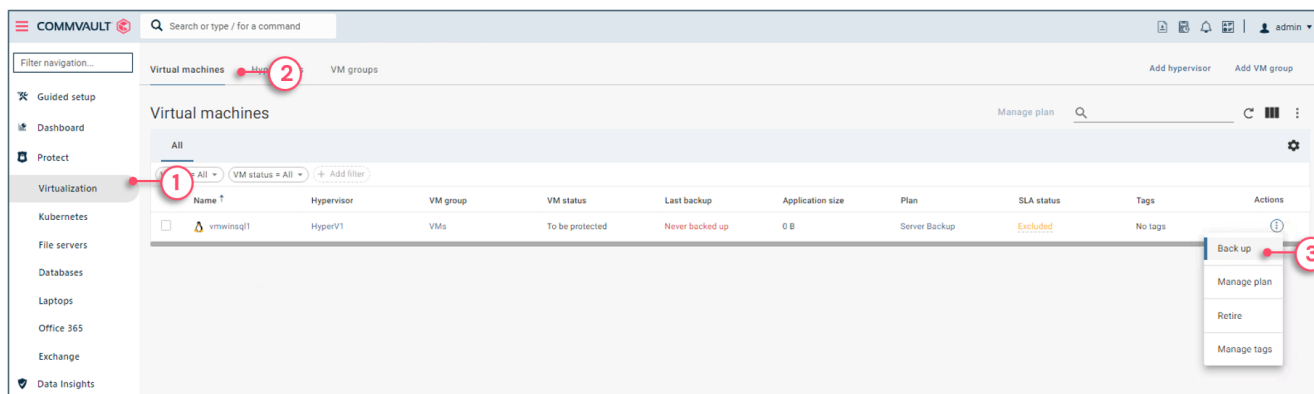
- From the VM Group view – Initiates a backup for all the VMs defined in the VM group/subclient
- From the virtual machine – Initiates a backup only for the selected virtual machine

A manual backup can be initiated for a VSA subclient, also called VM group. This protects all the virtual machines defined in the subclient. Use this method to resubmit a backup job that has failed but is not suited for an on-demand protection of a single VM.

Backup from a VM Group

1. Select **Protect > Virtualization**
2. Click the **VM groups** tab
3. Click the **Actions** icon for the desired subclient and then click **Backup**. You will then be able to choose the backup type.

Backup from a Virtual Machine



Virtual machines must be backed up at least once before you can request a backup on demand. An on-demand VM backup is always an incremental backup. If a VM is included in multiple subclients, an on-demand backup is run for the subclient that performed the latest backup of the VM.

1. Select **Protect > Virtualization**
2. Click the **Virtual machines** tab
3. Click the **Actions** icon and then click **Back up** to run a backup for the virtual machine; a manual single VM backup only allows an incremental backup

Recover a Full VM

The image consists of two screenshots from the Commvault user interface, illustrating the steps to recover a full VM.

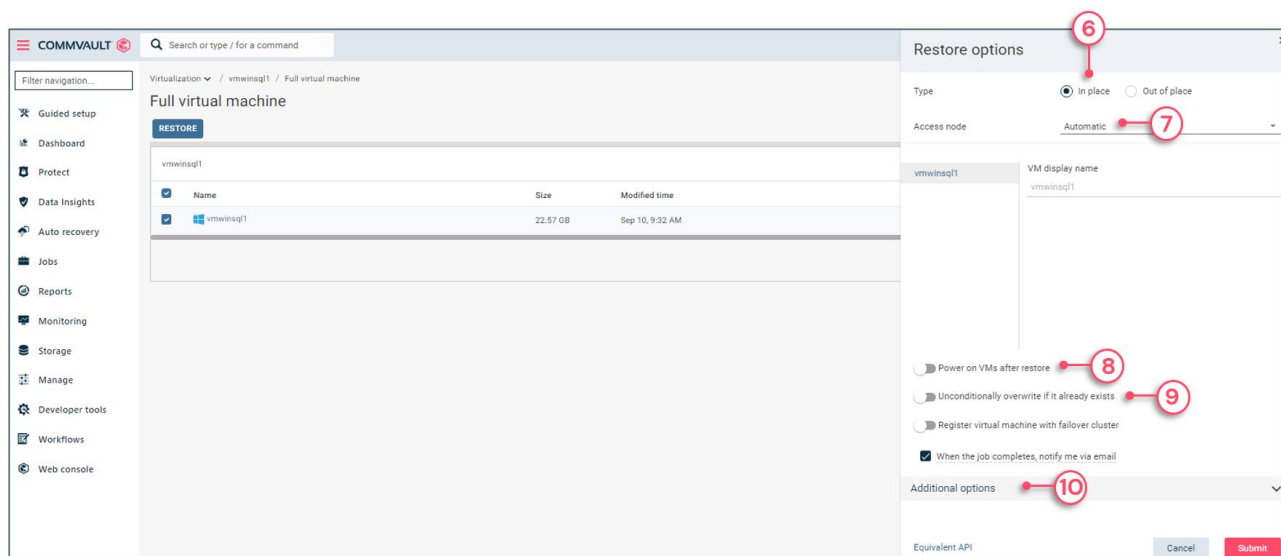
Top Screenshot: Shows the 'Virtual machines' page for 'vmwinq1'. The left sidebar has 'Virtualization' highlighted (1). The 'Virtual machines' tab is selected (2). The VM 'vmwinq1' is selected (3). The 'Recovery points' calendar shows a date (4) and a 'RESTORE' button.

Bottom Screenshot: Shows the 'Select restore type' dialog. The 'Full virtual machine' option is selected (5).

A virtual machine can be fully recovered in the same virtual environment or a different environment accessible by a VSA agent. The VM can be restored with the same name or can be renamed to avoid overriding the source virtual machine.

1. Select **Protect > Virtualization**
2. Click the **Virtual machines** tab
3. Click the name of the VM you wish to protect
4. From the **Overview** for the virtual machine, select the desired recovery point, then click **RESTORE**
5. Click **Full virtual machine** to recover the full VM

In Place Restore



6. *In place* is the default restore type
7. Choose the VSA access node that will execute the restore
8. Toggle on **Power on VMs after restore** to power on the VM upon restore
9. Toggle on **Unconditionally overwrite if it already exists** if you want to overwrite the existing VM
10. In the *Additional Options* dropdown list, choose disk provisioning type for the VM being restored, then allow the system to use the best available transport mode or choose the mode you want

Out of Place Restore

The screenshot displays the Commvault web interface for configuring a 'Full virtual machine' restore. The left sidebar contains navigation links such as 'Guided setup', 'Dashboard', 'Protect', 'Data Insights', 'Auto recovery', 'Jobs', 'Reports', 'Monitoring', 'Storage', 'Manage', 'Developer tools', 'Workflows', and 'Web console'. The main content area shows a table of VMs to be restored, with columns for 'Name', 'Size', and 'Modified time'. The right-hand panel contains configuration options for the restore process, including 'Destination client', 'Access node', 'Disk provisioning', 'VM display name', 'Destination HyperV host', 'Restore location', 'Hyper-V default folder', 'Destination folder', 'Network adapter', and various toggle options for power and overwrite settings. The 'Submit' button is highlighted with a red circle and the number 8.

Name	Size	Modified time
vmwinstq1	22.57 GB	Sep 10, 9:32 AM

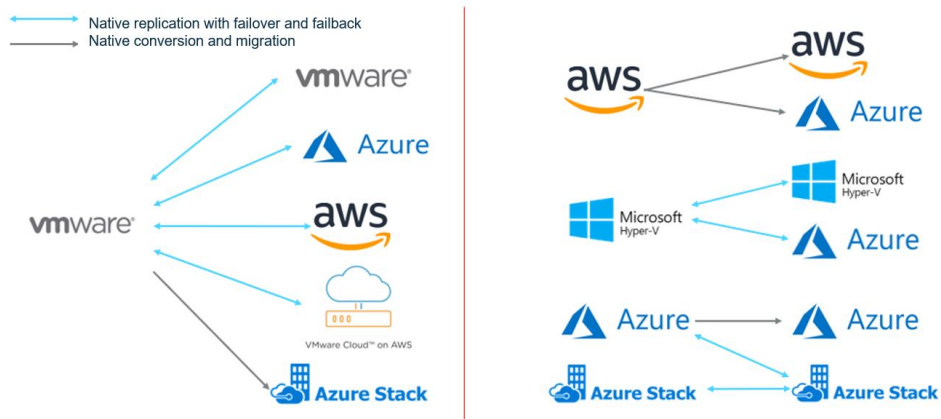
Configuration options on the right:

- Destination client: HyperV1 (1)
- Access node: Automatic (2)
- Disk provisioning: Auto
- VM display name: vmwinstq1-restore (3)
- Destination HyperV host: hyperv1
- Restore location: Hyper-V default folder (4)
- Destination folder: (Browse button)
- Network adapter: Not Connected (5)
- Power on VMs after restore (6)
- Unconditionally overwrite if it already exists (7)
- Register virtual machine with failover cluster
- When the job completes, notify me via email (checked)
- Additional options: (Expand arrow)
- Equivalent API: (Cancel button, Submit button (8))

After choosing **Out of place restore**:

1. Choose the virtualization vendor and the hypervisor on which to restore the VM
2. Choose the VSA access node that will execute the restore
3. Rename the VM on restore if needed
4. Choose the datastore, resource pool, and folder to use for recovery
5. Scroll to display additional restore option; choose to add or modify network and IP settings
6. Toggle on **Power on VMs after restore** to power on the VM after restore
7. Toggle on **Unconditionally overwrite if it already exists** if you want to overwrite the existing VM
8. Click **Submit** to start the recovery

Cross-Platform Recovery

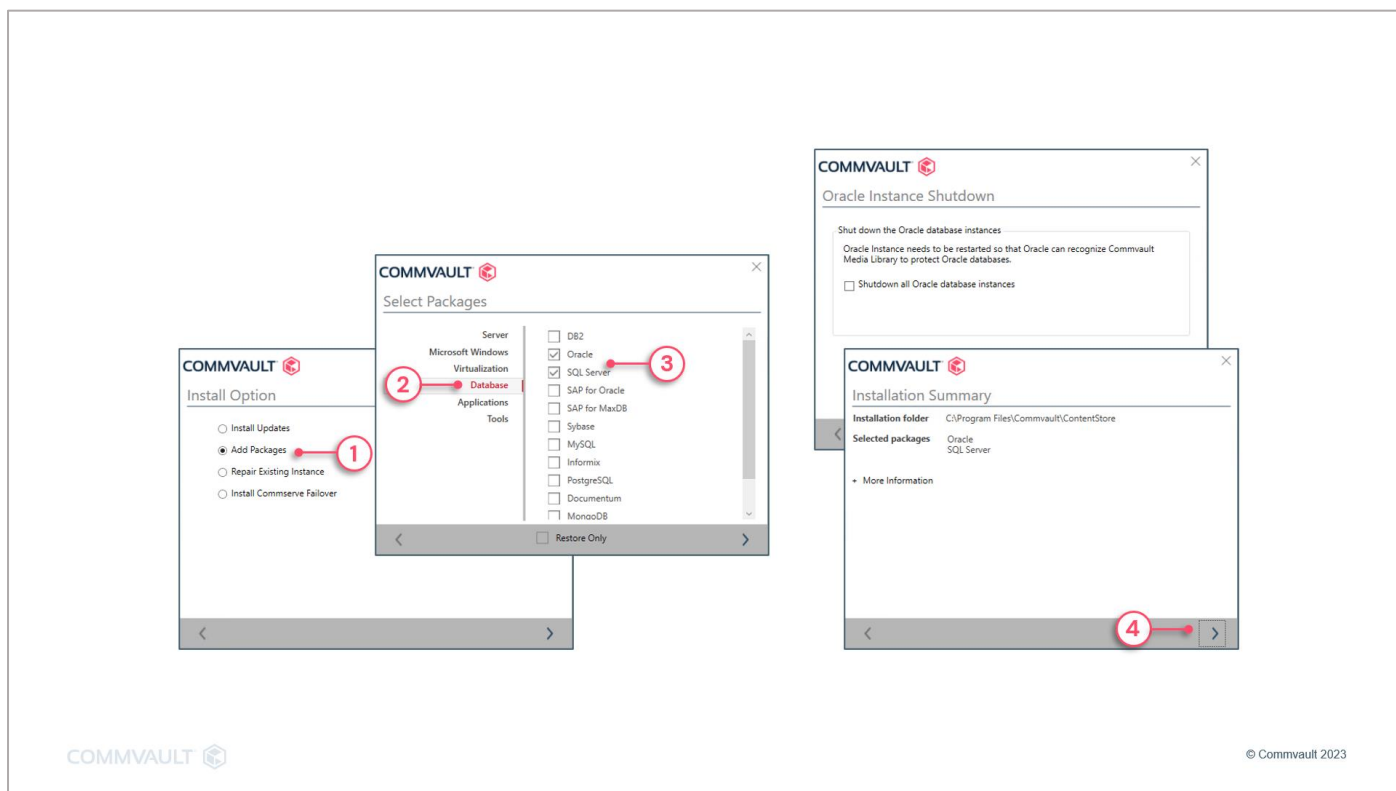


Backup and restore is not platform-dependent. Restores can be done between hypervisors with a conversion happening during the restore process. This feature can be used as a migration tool or to provide recovery capability between two data centers using different hypervisors. There are some limitations on the source and destination so refer to the Commvault Online Documentation for the latest matrix of source and destination hypervisors.

NOTE: Cloud virtual machines are configured as generic hypervisors and act as a source or target to the virtual server agent. Conversion occurs automatically during the restore process when the destination hypervisor is selected.

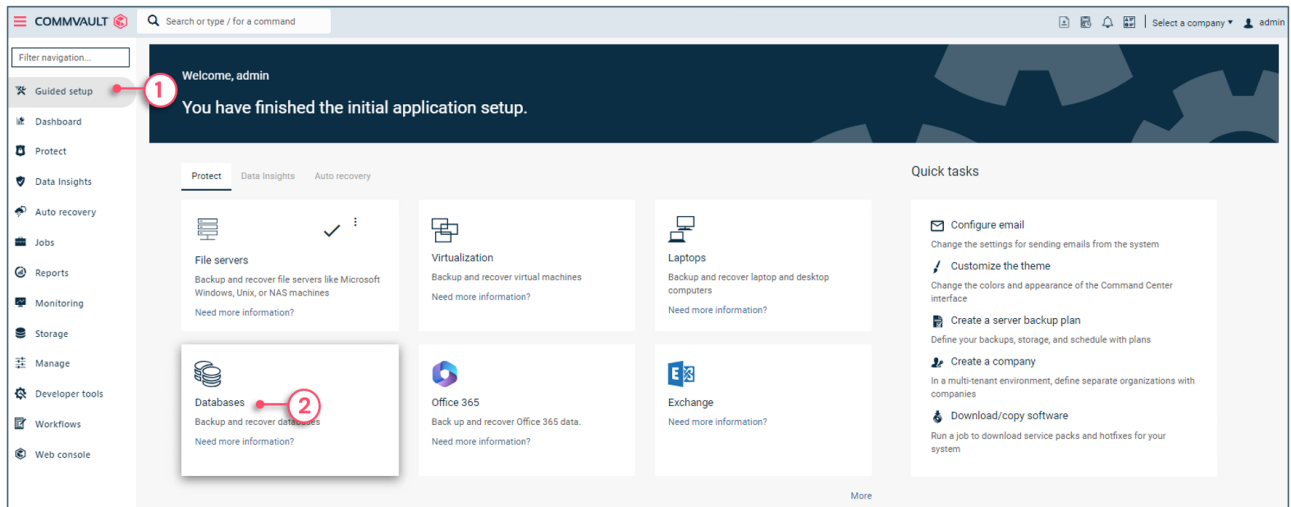
Database Protection

Add a Database Agent



1. Launch the **Setup.exe** and select **Add Packages**
2. Select the **Database package**
3. Select the database types
4. Click **>** to begin the installation

Configure the Database Feature



Database Protection

Commvault provides a single solution to protect data in multiple environments, providing a unified approach to administration, plan management, and data access. From dev/test to production to archiving and retirement, Commvault software covers the entire lifecycle of your enterprise's most important data, thereby improving application performance, accelerating innovation, reducing costs, and bringing copy data management under control.

The Commvault Command Center is configured to protect a variety of databases that run on physical servers, virtual machines, or cloud instances. Features like database snapshots, replication between two systems, database cloning with data masking, and restore to a sequence number or time interval are supported across a variety of database engines.

Database backups are performed the same way file system or virtual machines backups are executed. A subclient is defined and backups are scheduled or run manually. Administrators can define and perform backup procedures in the Command Center for a variety of databases without having to create scripts, special functions, or specific processes and procedures. Scripting for a specific database or database type is not needed.

Databases are protected by installing a database agent on the client that is running the database or on a MediaAgent that can connect to the database using the vendor-supported database connections. The database agent can connect to the database instance and copy the whole instance, the transaction logs, and individual tables.

Configure the Database Feature

1. Select **Guided setup**
2. Click the **Databases** tile

Configure the Database Feature ...cont'd

The screenshot shows the 'Databases setup' page in the Commvault web console. The 'Add database servers' form is displayed with the following fields and callouts:

- 3**: Application type (SQL server)
- 4**: OS Type (Windows selected)
- 5**: Database server name (vmwinaql1.cvlab)
- 6**: User name (cvladadministrator)
- 7**: Plan (Production Server plan)
- 8**: Add button

Other visible fields include Password, Installation location (Optional), and SQL server authentication (Optional). The 'Do it later' button is also present.

3. Choose a database engine
4. Choose the platform hosting the database
5. Enter the server's full name or IP address
6. Enter credentials with administrative privileges on the server
7. Choose a backup plan
8. Click **ADD** to continue; a task is created to install the database agent to the defined server

Configure a Database Instance

The screenshot displays the Commvault console interface. On the left, a navigation sidebar lists various categories, with 'Databases' highlighted and marked with a red circle and the number 1. The main content area shows the 'Instances' tab, which includes a summary bar (0 Met, 0 Missed, 1 Excluded, 0 May miss SLA) and a table of database instances. The table has columns for Name, Server, Database engine, Status, Last backup, and Application. One instance, 'VMWINSQL1', is listed with status 'Not Ready'. In the top right corner, an 'Add instance' button is marked with a red circle and the number 2. A modal dialog titled 'Add instance' is open on the right, showing a grid of database vendor logos including SQL server, Oracle, Oracle RAC, SQL availability group, SAP for Oracle, SAP HANA, DB2, DB2 MultiNode, PostgreSQL, MySQL, Informix, and Sybase. This dialog is outlined with a red border and marked with a red circle and the number 3.

1. Select **Databases**

2. Click **Add instance** to add a database instance

3. Select the database vendor

Configure Database Agent Security

The first screenshot shows the Commvault console interface. The left sidebar has a 'Databases' link highlighted with a red circle and the number 1. The main area shows a table of instances. The row for 'VMWINSQL1' is highlighted with a red circle and the number 3. The 'Status' column for this instance shows 'Not Ready' with a red circle and the number 2. The 'Last backup' column shows 'Never backed up'.

The second screenshot shows the 'Edit SQL Server authentication' dialog box. The 'Override higher levels settings' toggle is turned on. The 'Use local system account' radio button is selected. The 'Credentials' field is set to 'SQL Admin'. The 'Equivalent API' field is empty. The 'Edit' button is highlighted with a red circle and the number 4.

An important step in configuring a database agent is to define a security account to use to connect to the database instances. A good example is the Microsoft SQL agent that traditionally used a local system account to backup SQL databases. Current versions of MSSQL do not allow local accounts to be used and therefore require the agent to be configured with a service account with the proper permissions.

1. Select **Protect > Databases**
2. If an agent security is not properly configured, the status will show *Not Ready*
3. Click the server's name to configure the security
4. In the *General* section, click **Edit**

Backup a Database Instance

The screenshot displays the Commvault console interface. On the left, the 'Databases' option is selected in the navigation menu (1). The main area shows a table of instances. The first instance, 'VMWINSQL1', is selected (2). The 'Actions' icon for this instance is clicked (3), opening a dropdown menu. The 'Back up' option is selected from this menu. A modal window titled 'Select backup level' is then displayed, where the 'Full' backup level is selected (4). The modal also shows 'Advanced options' and 'Equivalent API' at the bottom.

1. Select **Protect > Databases**
2. For the desired server, click the **Actions** icon
3. Select **Back up**
4. Select backup type

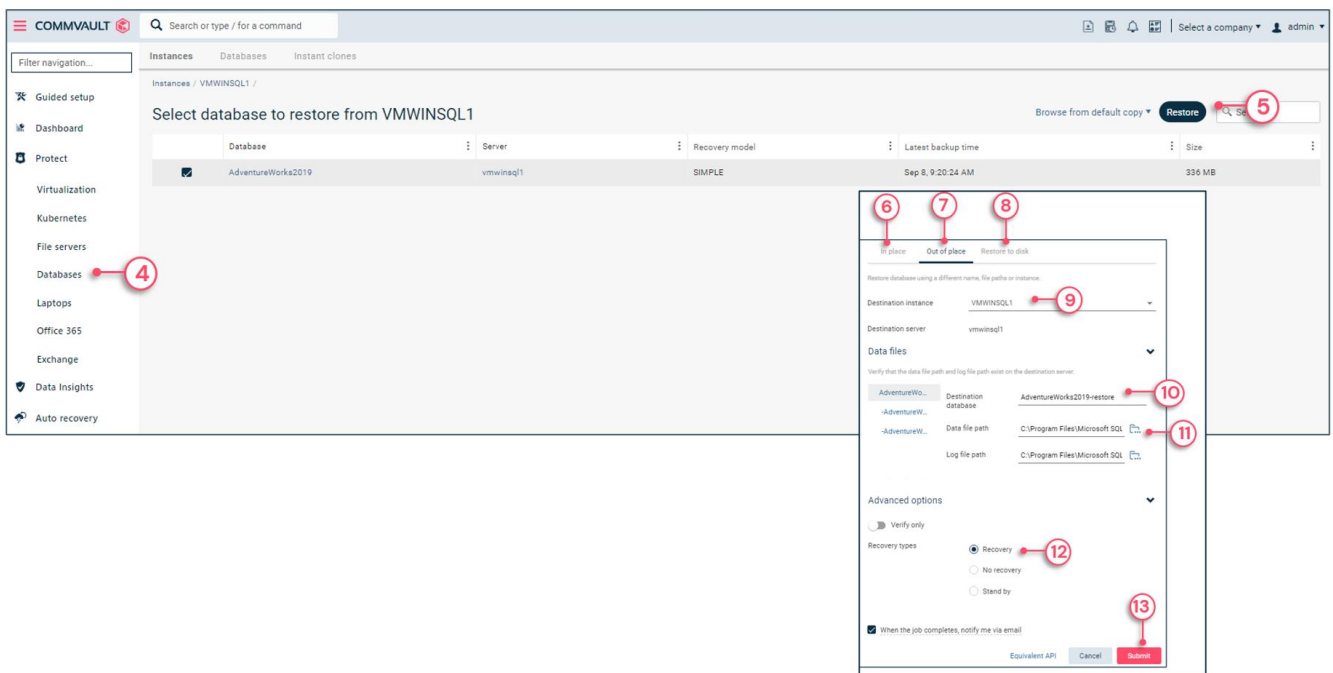
Database Recovery

The screenshot displays the Commvault web interface for database recovery. On the left, a sidebar lists various protection categories, with 'Databases' highlighted. The main content area shows the 'VMWINSQL1' instance overview. A red circle '1' points to the 'General' tab, which lists instance details. Another red circle '2' points to the 'Instances' header. A third red circle '3' points to the 'Recovery points' section, which includes a calendar and a timeline. A red box highlights the 'Restore' button at the bottom right of the Recovery points section.

Databases can be recovered using a variety of methods but must be recovered back into a similar database.

1. Select **Protect > Databases**
2. Navigate to the instance you wish to recover
3. From *Overview* of the required instance, select the required recovery point, then click **Restore**

Database Recovery ...cont'd



4. Select the database to restore
5. Click **Restore**
6. Select **In place** to restore the database in place and overwrite the existing production database
7. Select **Out of place** to restore the database to another location
8. Select **Restore to disk** to restore the database to a disk
9. Choose the instance name in which to recover the database if using out of place restore
10. Enter a name for the destination database
11. Enter the location for the database files and log files
12. Choose the recovery type
13. Click **Submit**